



QUESTION BANK

FUNDAMENTALS OF QUANTUM COMPUTING

UNIT I

Short questions

1. Define Rayleigh-jeans law?

According to Rayleigh-jean's the radiation emitted by the black body at a temperature is directly proportional to absolute temperature of black body and inversely proportional to fourth power of the wavelength.

$$E \propto T$$
$$E(u) \propto \frac{1}{\lambda^4}$$
$$E(u) = \frac{8\pi kT}{\lambda^4}$$

2. Define Stephen-Boltzmann law?

Stephen-Boltzmann law is the relation between the radiation emitted by the black body to the temperature.

The total radiation emitted by the black body is directly proportional to the fourth power of absolute temperature of black body.

$$\text{i.e., } E \propto T^4$$

$$E = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8} \text{ w/m}^2 - \text{K}^4.$$

3. Write the expression for the Weins law?

It explains a shift of peak wavelength with the temperature.

The peak wavelength emitted by the black body temperature is inversely proportional to the absolute temperature of black body.

$$\text{i.e., } \lambda_m \propto \frac{1}{T}$$
$$\lambda_m = 2.87 \times 10^{-3} \frac{mK}{T}$$



4. Define black body radiation?

A perfect black body one which observed and emit all the wavelength of incident radiation on it. The radiation emitted by the black body is known as Black body radiation.

5. Define Matter waves?

Any moving particle is associated with wave nature those waves are called matter waves.

Matter waves are not mechanical waves and electromagnetic wave these are new kind of waves proposed by De-Broglie.

Wavelength $\lambda = h/mv$

6. Explain the Heisenberg uncertainty principle?

According to Heisenberg it is difficult to find the position and momentum simultaneously.

x is the exact position of a particle 'p' is the exact momentum of a particle.

Δx is the error to find the position of a particle, ΔP is the error to find the momentum of a particle. Then the relation between Δx & ΔP from the Heisenberg can be written as

$$\Delta x \cdot \Delta P \geq \frac{h}{4\pi} \vee \Delta x \cdot \Delta P \geq \frac{h}{2}$$
$$\therefore h = \frac{h}{2\pi}$$

7. What particle and write down its characteristics?

A particle is a small, localized object in physics that can be treated as a single point in space with mass and other physical properties.

Characteristics:

Has mass and may carry charge.

Occupies negligible space (considered a point in ideal cases).



8. What is the statement of Energy quantization by max Planck?

Max Planck proposed that energy is not emitted or absorbed continuously, but in discrete packets called "quanta." According to his theory, the energy E of each quantum is directly proportional to the frequency ν of the radiation, given by the equation $E = h\nu$. This concept explained blackbody radiation and marked the beginning of quantum theory, introducing the idea that atomic and subatomic processes involve quantized energy levels.

9. Write down the types of radiation?

Radiation can be classified into several types based on its nature and how it interacts with matter. The main types of radiation are:

1. Ionizing Radiation

These have enough energy to ionize atoms (remove tightly bound electrons).

Alpha (α) radiation, Beta (β) radiation, Gamma (γ) radiation, X-rays, Neutron radiation

2. Non-ionizing Radiation

These do not carry enough energy to ionize atoms but can excite them or cause heating, Ultraviolet (UV) radiation, Visible light, Microwaves, Infrared (IR) radiation, Radio waves.

10. What is the need for quantum theory?

Quantum theory is needed to explain phenomena that classical physics cannot, such as blackbody radiation, the photoelectric effect, and atomic structure. It accurately describes the behavior of particles at atomic and subatomic scales, where energy is quantized and particles exhibit wave-particle duality, leading to advances in modern physics and technology.

Long questions

1. Explain planck's radiation law?

- Energy emitted or observed is due to vibration is due to vibration of micro particle (e).
- According to Max-Planck the energy radiated in the form of Quanta. The energy of Quanta $E = h\nu$ in discrete manner.
- The emitted energy is integer multiple of energy of Quanta ($E = h\nu$).

Let us consider black body with large no. of particles N ($N_0, N_1, N_2, N_3, \dots$) with discrete energies E ($0E, 1E, 2E, 3E, \dots$)



$N(N_0, N_1, N_2, N_3, \dots)$

$E(0E, 1E, 2E, 3E, \dots)$

Total no. of Black body

$$N = N_0 + N_1 + N_2 + N_3 + \dots \quad (1)$$

Boltzmann relation

$$N_r = N_0 e^{-rE/KT}$$

$$r = 0, 1, 2, 3, \dots$$

By substituting "r" value in

$$r = 0$$

$$N_0 = N_0 e^{-0.E/KT}$$

$$N_0(1)$$

$$N_0 = N_0$$

$$r = 1$$

$$N_1 = N_0 e^{-1E}$$

$$N_0 e^{-E/KT}$$

$$r = 2$$

$$N_2 = N_0 e^{-2E/KT}$$

$N_0, N_1, N_2, N_3, \dots$ in equation (1)

$$N = N_0 + N_0 e^{-E/KT} + N_0 e^{-2E/KT} + N_0 e^{-3E/KT} + \dots$$

$$= N_0 [1 + e^{-E/KT} + e^{-2E/KT} + e^{-3E/KT} + \dots]$$

$$\text{let } x = e^{-E/KT}$$

$$N_0 [1 + x + x^2 + x^3 + \dots]$$

$$\therefore (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$N = N_0 (1-x)^{-1}$$

$$N = \frac{N_0}{(1-x)} \quad (2)$$

Energy of the system

$$E = N_0(0E) + N_1(1E) + N_2(2E) + N_3(3E) + \dots$$

$$N_1(1E) + N_2(2E) + N_3(3E) \quad (3)$$

Substitute N_1, N_2, N_3 values from above



$$N_0 e^{-E/KT} (E) + N_0 e^{-2E/KT} (2E) + N_0 e^{-3E/KT} (3E) + \dots$$

$$N_0 e^{-E/KT} (F) [1 + 2e^{-E/KT} + 3e^{-2E/KT} + 4e^{-3E/KT} + \dots]$$

$$\text{Let } x = e^{-E/KT}$$

$$= N_0 x E [1 + 2x + 3x^2 + 4x^3 + \dots]$$

$$[\because (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots]$$

$$= N_0 x E (1-x)^{-2}$$

$$E = \frac{N_0 x E}{(1-x)^2}$$

4

$$\text{Average energy of system} = \frac{\text{Total energy of system}}{\text{Total no. of particles}}$$

$$\bar{E} = \frac{\frac{N_0 x E}{(1-x)^2}}{\frac{N_0}{(1-x)}}$$

$$\bar{E} = \frac{N_0 x E}{(1-x)^2} \times \frac{1-x}{N_0}$$

$$\bar{E} = \frac{x E}{(1-x)}$$

$$\bar{E} = \frac{E}{x^{-1} - 1}$$

$$\therefore x = e^{-E/KT}$$

$$\bar{E} = \frac{E}{(e^{h\nu/KT} - 1)} [\because E = h\nu]$$

$$\bar{E} = \frac{h\nu}{e^{h\nu/KT} - 1} \quad \text{5}$$

Black body absorbed and emit all the wavelength (from minimum wavelength to maximum) in the range λ to $\lambda + d\lambda$.

The no. of photons per unit volume

$$\frac{8\pi}{\lambda^4} d\lambda \quad \text{6}$$

By multiplying equation 5 with 6 we can get total energy emitted or absorbed by black body.



$$E d\lambda = \frac{h\nu}{(e h\nu/KT - 1) \times \frac{8\pi}{\lambda^4} d\lambda}$$

Equation 7 is known as Planck's law for Black body radiation. It explained blackbody radiation for every wavelength.

2. What is black body radiation? Write a note on laws of Black body radiation?

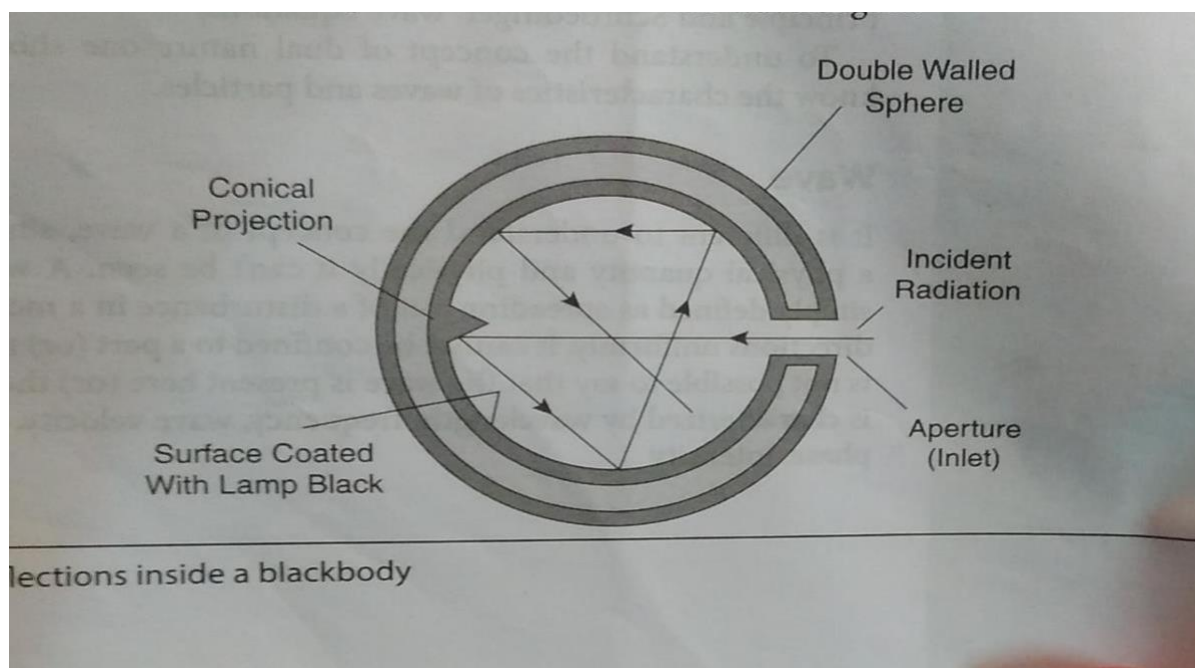
Black body:

A perfect black body one which observed and emitted all the wavelength of incident radiation on it.

Black body radiation:

The radiation emitted by the black body is known as Black body radiation.

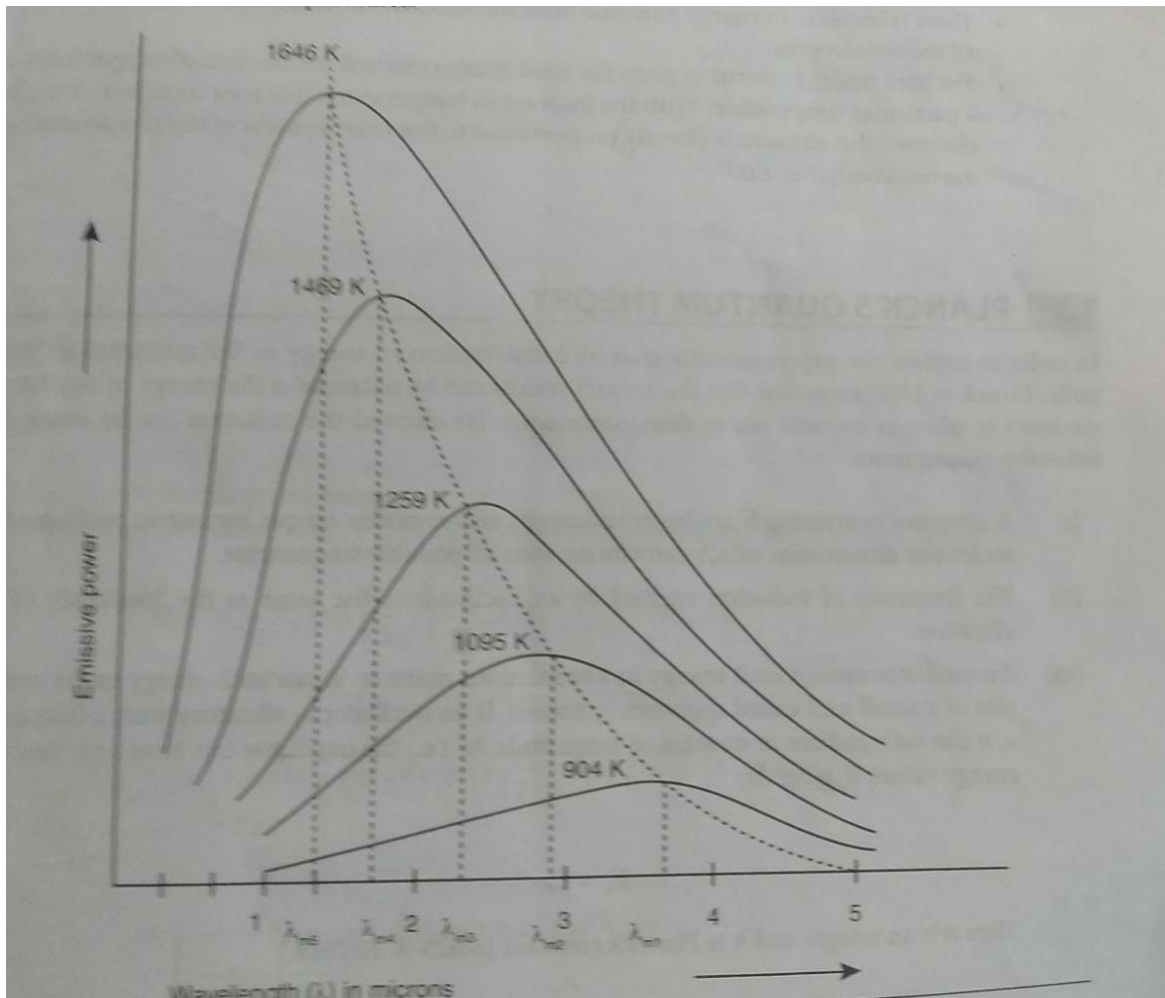
Construction of Black body: Black body is a copper hallow sphere its inner surface is coated with the lamp black; a hole is made on it for entering the radiation. When the radiation entered through this hole into copper hallow sphere due to multiple reflections all the wavelength of radiation is observed. When the copper hallow sphere heated to a particular temperature (or) at thermal equilibrium it will emit all the wavelength of radiation.





Energy spectrum for Black body radiation:

By plotting the graph between wavelength and energy distribution of black body it will be



Energy distribution is not uniform at any given temperature. At a given temperature, with increasing wavelength intensity distribution of black body increases at a particular wavelength reaches maximum (λ_m) behind that intensity decreases.

With increase in temperature (λ_m) (maximum radiation emitted by the black body) decreases. The area of the curve at a temperature will give the radiation emitted by the black body.



Stephen - Boltzmann law:

Stephen-Boltzmann law is the relation between the radiation emitted by the black body to the temperature.

The total radiation emitted by the black body is directly proportional to the fourth power of absolute temperature of black body.

$$\text{i.e., } E \propto T^4$$

$$E = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8} \text{ w/m}^2 - \text{K}^4.$$

Wein's law (Wein's displacement law):

- It explains the shifting of peak wavelength with the temperature.
- The peak wavelength emitted by the black body temperature is inversely proportional to the absolute temperature of black body.

$$\lambda_m \propto \frac{1}{T}.$$
$$\lambda_m = 2.87 \times 10^{-3} \frac{\text{mK}}{T}.$$

Rayleigh-Jean's law

According to Rayleigh-jean's the radiation emitted by the black body at a temperature is directly proportional to absolute temperature of black body and inversely proportional to fourth power of the wavelength.

$$E \propto T$$
$$E(u) \propto \frac{1}{\lambda^4}$$
$$E(u) = \frac{8\pi kT}{\lambda^4}$$



3.Explain de-broglie Hypothesis?

De-Broglie proposed the dual nature for the matter. According to De-Broglie matter exhibit both nature particle nature and wave nature.

Matter wave:

Any moving particle is associated with wave nature those waves are called matter waves.

Matter waves are not mechanical wave and electromagnetic wave these are new kind of waves proposed by De-Broglie.

Wave length of matter wave $\lambda = \frac{h}{mv} = \frac{h}{p}$

h = Planck's constant.

$h = 6.825 \times 10^{-34}$ J. Sec, 6.62×10^{-27} erg-sec.

m = Mass of the particle.

v = Velocity of the particle.

De-Broglie wavelength for the matter wave:

De-Broglie calculated the wavelength of matter wave by combining Max-Planck's radiation law and Einstein Mass-Energy relation.

According to Max-Planck's radiation law $E = h\nu$ ————— ①

According to Einstein mass-energy relation $E = mc^2$. ————— ②

From equation 1 and 2

$$h\nu = mc^2 \quad \left[\because u = \frac{c}{\lambda} \right]$$

$$\frac{hc}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mc} \quad c = 3 \times 10^8 \text{ m/s}$$

Particle velocity ' v '

$$\text{The } \lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{p}$$

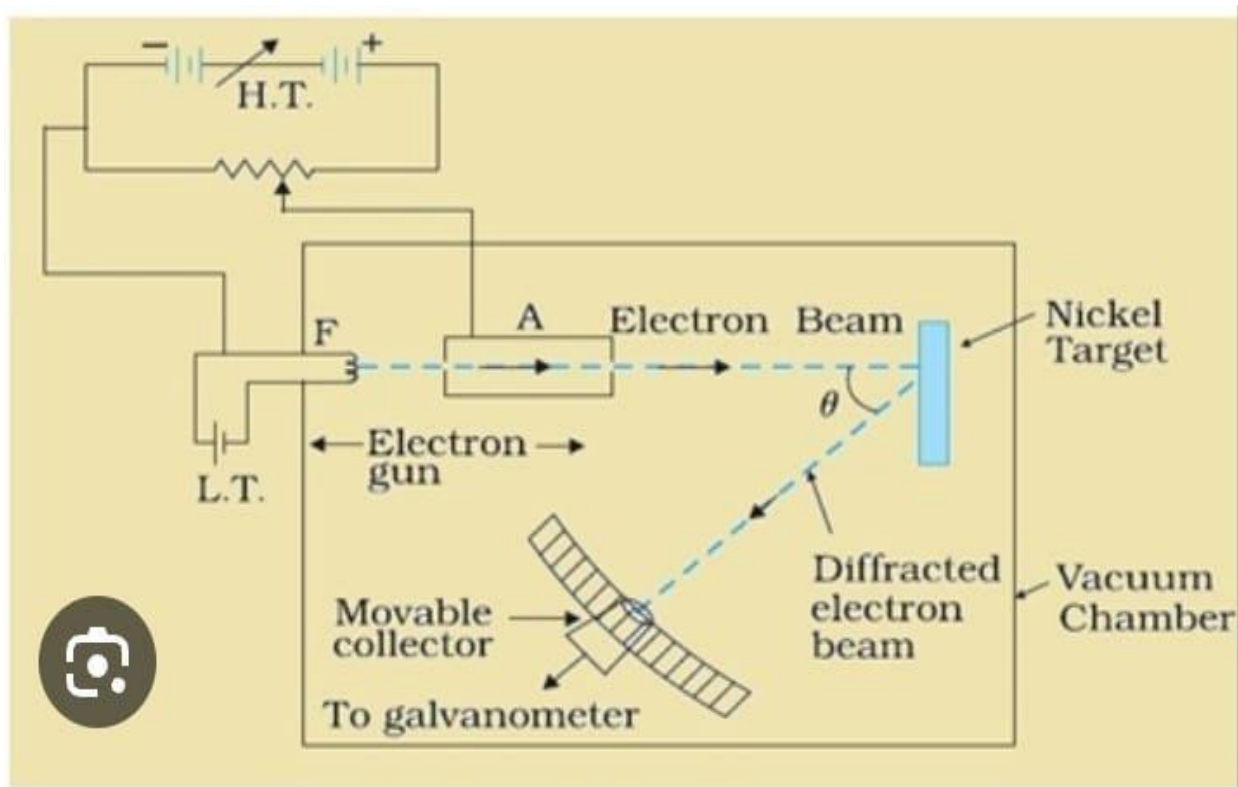
4.Describe the experiment to verify the existence of matter waves?

Davisson and Germer experiment is the practical evidence for the wave nature of the matter.



This was explained by CH. Davisson and LH. Germer

The experimental set up of Davisson and Germer experiment is



1) Electron gun:

Electron gun produces Electron beam it consists of filaments, low tension battery, High tension battery, pin hole cylindrical arrangement to get fine single beam of electron.

When filament gets heated electron gun produces electron beam these electrons are accelerated by providing sufficient potential from High tension batteries.

2) Target set-up (Ni-crystal):

The fast movement of electron beam from electron gun is made incident on Ni-crystal, due to atomic planes of Ni-crystal electrons get diffracted in all possible directions.

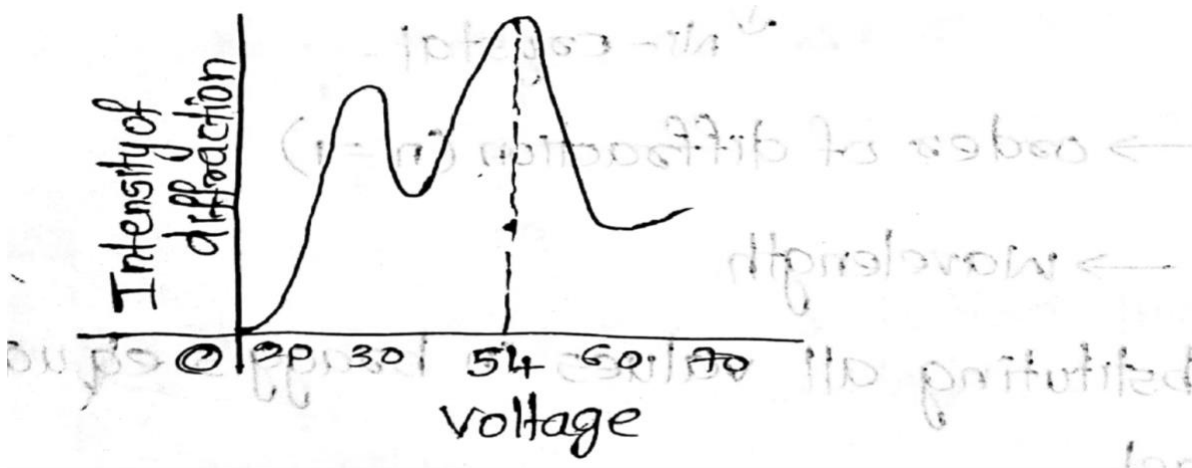
3) Circular scale arrangement:



It is used to collect the diffracted electron beam. It consists of one electron collector which is movable on circular scale, the electron collector is connected to sensitive galvanometer to measure the intensity of diffracted electron beam.

Calculation of wavelength associate from the electron:

When the voltage from High tension battery $V = 54$, The angle between incident beam to diffracted beam is 50° then we will get maximum diffraction pattern (the order of diffraction $n=1$).



The wavelength associate from the electron according to de-broglie

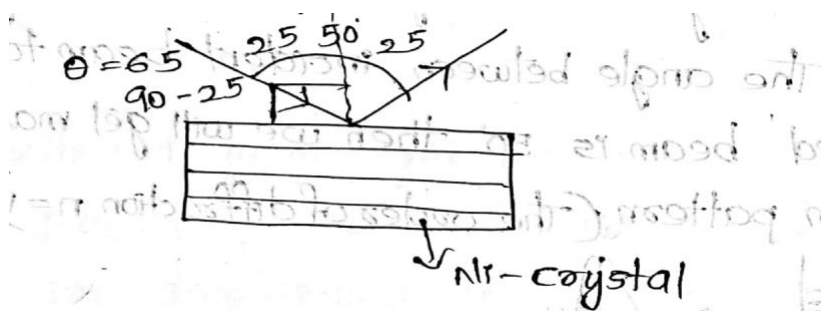
$$\lambda = \frac{12.26}{\sqrt{V}} \text{Å}$$

$$\lambda = \frac{12.26}{\sqrt{54}}$$

$$\lambda = 1.65 \text{Å}$$

The wavelength is associated from Bragg's law for the electron.

Bragg's law $2d\sin\theta = n\lambda$



By substituting all values in Bragg's equation, we get



$$2 \times 0.91 \times 10^{-10} \sin 65^\circ = 1 \times \lambda$$

$$\lambda = 1.66 \text{ \AA}$$

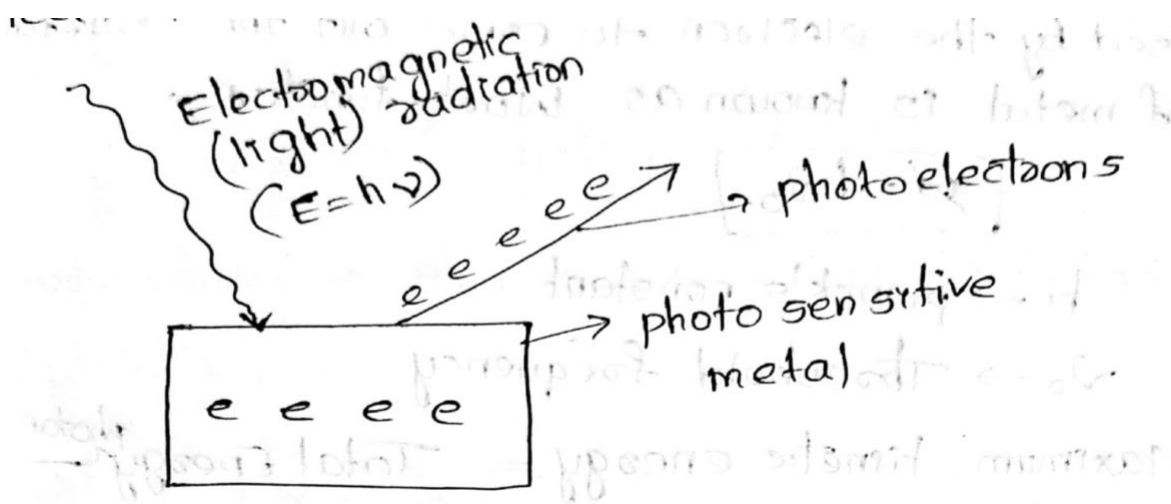
A good agreement between De-Broglie and Bragg's law for the electron

Hence, the wave nature of electron was proved.

5. Define photoelectric effect? With mathematical expression?

When the electromagnetic radiation (light) is incident on a photo sensitive metal if the electrons are emitted/ejected from the surface of metal then that phenomenon is called photo electric effect.

Einstein explained photo electric effect phenomena by using Max-Planck quantum theory.



According to Einstein, energy radiated in the form of photon.

Energy of photon $E = h\nu$

Explanation:

When the electromagnetic radiation is incident on a photosensitive metal photon ($E = h\nu$) collides with the electron, during this collision photon transfer all the amount of energy to electron.

In these total energy part of energy used by the electron comes out the surface of the metal remaining part of energy used by the electron as maximum Kinetic energy with this electron leaves from the metal surface.

Work function:

The minimum amount of energy used by the electron to come out the surface



of metal is known as work function. $\phi = h\nu_0$

$h \rightarrow$ Plank's constant.

$\nu_0 \rightarrow$ Threshold frequency.

Maximum Kinetic energy = Total energy photon - Work function.

$$KE = h\nu - \phi$$

$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

$$\frac{1}{2}mv^2 = h(\nu - \nu_0)$$

Condition for photo electric effect:

We know mathematical expressions for photo electric effect.

$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

Kinetic energy is either positive (or) zero.

i.e., $K.E \geq 0$

$$\rightarrow h\nu - h\nu_0 \geq 0$$

$$h\nu \geq h\nu_0$$

$$\nu \geq \nu_0$$

$\nu \rightarrow$ Incident radiation frequency.

$\nu_0 \rightarrow$ Threshold frequency.

Conclusion:

To observe the photo electric effect phenomena incident radiation frequency must be greater than (or) equal to threshold frequency.

6. . Write a note on Born's interpretation of wave function. Derive the time-dependent Schrodinger's wave equation of electron?

Wave function is a mathematical expression represented in complex number. It has no direct physical meaning.

square of the wave function $[\psi^2(x) = \psi(x)\psi(x)]$

is called probability density

It gives finding the position of a particle in a given region.

If the particle is presented in a given region, then probability density



maximum this condition is known as normalization Condition.

Mathematically we can represent as.

$$\int \psi^2(x) d(x) = 1[1 - D]$$

$$\int \psi^2(x, y, z) dv = 1[3 - D]$$

If the particle does not exist in a given region, then probability density

minimum this condition is known as orthogonal condition.

Mathematically represented as.

$$\int \psi^2(x) dx = 0[1 - D]$$

$$\int \psi^2(x, y, z) dv = 0[3 - D]$$



This is the Schrodinger Time Independent Wave Equation

Schrodinger's time dependent wave equation.

$$\nabla \cdot \nabla \psi + \frac{2m}{\hbar^2} [E - V] \psi = 0$$

Multiply on both sides $\hbar^2/2m$

$$\hbar^2/2m [\nabla^2 \psi + \frac{2m}{\hbar^2} [E - V] \psi] = 0 \times \hbar^2/2m$$

$$\frac{\hbar^2}{2m} \times \nabla^2 \psi + \frac{\hbar^2}{2m} \times \frac{2m}{\hbar^2} [E - V] \psi = 0 \times \frac{\hbar^2}{2m}$$

$$\frac{\hbar^2}{2m} \times \nabla^2 \psi + [E - V] \psi = 0$$

$$E\psi = -\frac{\hbar^2}{2m} \times \nabla^2 \psi + V\psi$$

$$E\psi = \left[-\frac{\hbar^2}{2m} \times \nabla^2 + V \right] \psi$$

$$E\psi = H\psi \quad \left[H = -\frac{\hbar^2}{2m} \times \nabla^2 + V \right]$$

H is called Hamiltonian operator. E is called energy operator. This is known as Schrodinger's time dependent wave equation.

Note:- The Schrodinger wave equation describes the behaviour of matter waves.

7. Derive the time-independent Schrodinger's wave equation of electron ?

Schrodinger wave equation is the fundamental wave equation in Quantum physics to explain the motion of microscopic particles.

To derive the Schrodinger wave equation

Let us consider an electron with mass 'm' moving with velocity 'v' in positive 'x' direction under the potential 'V'.

The general wave equation for the electron can be represented as

$$\psi(x) = A \sin\left(\frac{2\pi}{\lambda}x\right) \quad \text{①}$$

Differentiate above with r to "x".

$$\frac{d\psi(x)}{dx^2} = A \cos\left(\frac{2\pi}{\lambda}x\right) \times \frac{2\pi}{\lambda}$$



Again differentiate above with respect to “x”

$$\frac{d^2\psi(x)}{dx^2} = -A\sin\left(\frac{2\pi}{\lambda}x\right) \cdot \left(\frac{2\pi}{\lambda}\right)\left(\frac{2\pi}{\lambda}\right)$$

$$\frac{d^2\psi(x)}{dx^2} = \frac{-4\pi^2}{\lambda^2} A\sin\left(\frac{2\pi}{\lambda}x\right)$$

From equation 1

$$\frac{d^2\psi(x)}{dx^2} = \frac{-4\pi^2}{\lambda^2} \psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\lambda^2} \psi(x) = 0 \text{ (2)}$$

From the De-Broglie

$$\therefore \lambda = \frac{h}{mv}$$

λ substitute in equation 2

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2}{\left(\frac{h}{mv}\right)^2} \psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2 m^2 v^2}{h^2} \psi(x) = 0 \text{ (3)}$$

Total energy of particle = K.E + P.E.

$$E = \frac{1}{2} mv^2 + V$$

$$E - V = \frac{1}{2} mv^2$$

$$2(E - V) = mv^2.$$

Multiply with “m” on both sides.

$$m \times 2(E - V) = m \times mv^2 \quad 2m(E - V) = m^2 v^2 \text{ substitute } \in \text{ eqn 3}$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{4\pi^2(2m(E - V))}{h^2} \psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2 m(E - V)}{h^2} \psi(x) = 0 \text{ (4)}$$

$\psi(x) \rightarrow$ Wave function.

$x \rightarrow$ Propagation of wave.



$m \rightarrow$ Mass of the particle.

$E \rightarrow$ Total Energy of the particle.

$V \rightarrow$ Applied potential.

$h \rightarrow$ Planck's constant.

Equation 4 is known as Schrodinger wave equation (or) Schrodinger 1-D wave equation (or) Time independent wave equation.

$$\text{Since } \therefore k = \frac{h}{2\pi}$$

$$\rightarrow h = k2\pi$$

Substitute in equation 4

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V)}{k^24\pi^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m(E - V)}{k^2}\psi(x) = 0$$

This is also Schrodinger wave equation in terms of 'h'

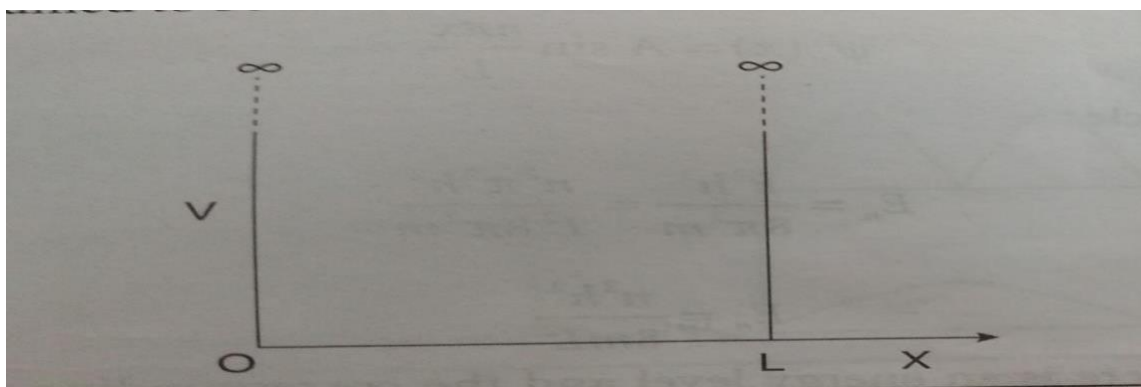
Schrodinger wave equation in 3-Dimension can be represented as

$$\frac{d^2\psi(x)}{dx^2} + \frac{d^2\psi(y)}{dy^2} + \frac{d^2\psi(z)}{dz^2} + \frac{8\pi^2m(E - V)}{h^2}\psi(x, y, z) = 0$$

8. Derive a wave function equation for electron by Trapping in one dimensional potential box?

Let us consider a one-dimensional potential box of width 'a'

Let consider an electron mass is 'm' moving with velocity v in positive x direction.





The motion of microscopic particle is explained by Schrodinger wave equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V(x))}{h^2}\psi(x) = 0 \quad \text{①}$$

Inside of the box

$$V(x) = 0; 0 < x < a$$

Outside of box

$$V(x) = \infty; x < 0()x > a$$

Apply Schrodinger wave equation inside box of

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - V(x))}{h^2}\psi(x) = 0 [\because V(x) = 0]$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m(E - 0)}{h^2}\psi(x) = 0$$

$$\frac{d^2\psi(x)}{dx^2} + \left(\frac{8\pi^2mE}{h^2}\right)\psi(x) = 0 [\because K^2 = \frac{8\pi^2mE}{h^2}]$$

$$\frac{d^2\psi(x)}{dx^2} + K^2 \psi(x) = 0 \quad \text{②}$$

The solution of equation 2 from the second order differentiation can be written as

$$\psi(x) = A\sin kx + B\cos kx \quad \text{③}$$

Boundary conditions for 1-D potential box.

At I-Boundary the wave function is 0

$$Atx = 0 \Rightarrow \psi(x) = 0$$

At II-Boundary the wave function 0

$$Atx = a \Rightarrow \psi(x) = 0$$

Apply Boundary condition I & II for equation 3

$$Atx = 0 \Rightarrow \psi(x) = 0 \psi(a) = A\sin kx + B\cos kx$$

$$0 = A\sin k(0) + B\cos(0)$$

$$\Rightarrow 0 + B = 0$$

$$B = 0$$

$$Atx = a \Rightarrow \psi(x) = 0$$



$$0 = A \sin k(a) + B \cos k(a)$$

$$A \sin ka + B \cos ka (\because B = 0)$$

$$\Rightarrow A \sin ka = 0$$

$$\sin ka = 0$$

$$ka = \sin^{-1}(0)$$

$$\sin^{-1}(\sin \pi) ka = n\pi.$$

$$k = \frac{n\pi}{a} [k \rightarrow \text{wavevector}]$$

$$\therefore k^2 = \frac{8\pi^2 m E}{h^2}$$

Substitute "k" value.

$$\left(\frac{n\pi}{a}\right)^2 = \frac{8\pi^2 m E}{h^2}$$

$$\frac{n^2 \pi}{a^2} = \frac{8\pi^2 m E}{h^2}$$



$$E_n = \frac{n^2 h^2}{8ma^2}$$

If $n=1$

$$E_1 = \frac{(1)^2 h^2}{8ma^2} = \frac{h^2}{8ma^2} \text{ (Ground state of electron)}$$

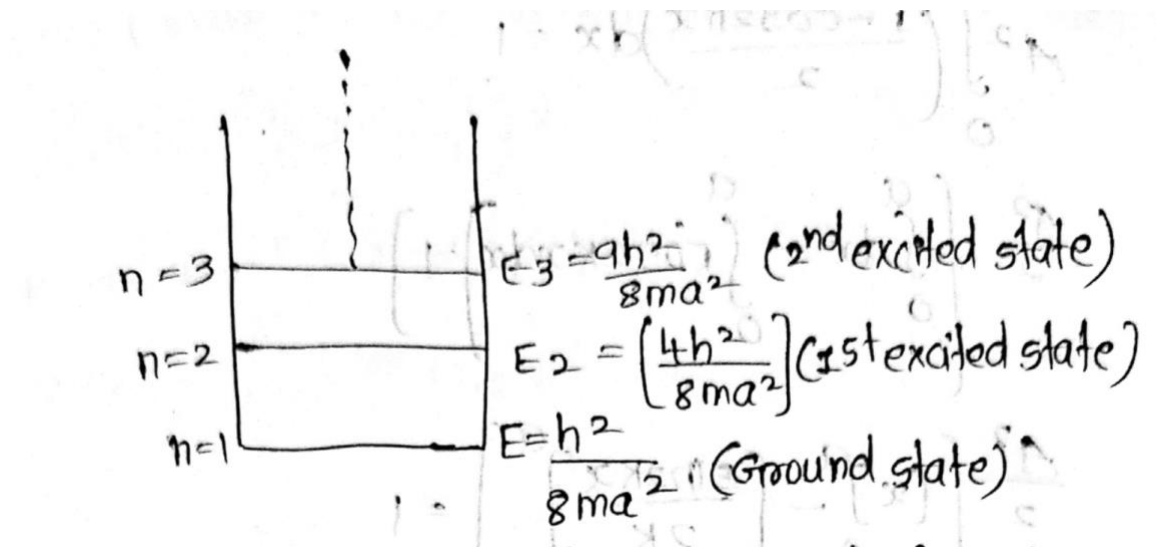
If $n=2$

$$E_2 = \frac{4h^2}{8ma^2} \text{ (1st excited state)}$$

If $n=3$

$$E_3 = \frac{9h^2}{8ma^2} \text{ (2nd excited state)}$$

Energy of electron is Quantized and emit in discrete manner.



Let us assume we can find the position of electron inside of potential box.

Normalization condition $\int \psi^2(x) dx = 1$

$$\int_0^a \psi^2(x) dx = 1$$

From equation 3

$$\psi(x) = A \sin kx - B \cos kx \quad [\because B = 0]$$

$$\psi(x) = A \sin kx$$

Substitute in above

$$\int_0^a (A \sin kx)^2 dx = 1$$



$$\int_0^a A^2 \sin^2 kx dx = 1$$

$$A^2 \int_0^a \sin^2 kx dx = 1$$

$$A^2 \int_0^a \left(\frac{1 - \cos 2kx}{2} \right) dx = 1$$

$$\frac{A^2}{2} \left[\int_0^a dx - \int_0^a \cos 2kx dx \right] = 1$$

$$\frac{A^2}{2} \left[[x]_0^a - \left[\frac{\sin 2kx}{2k} \right]_0^a \right] = 1$$

$$\frac{A^2}{2} \left[(a - 0) - \left[\frac{\sin 2\pi(a)}{a} - \frac{\sin 2n\pi(0)}{a} \right] \frac{2n\pi}{a} \right] = 1$$

$$\frac{A^2}{2} [a - 0] = 1$$

$$\frac{A^2}{2} a = 1$$

$$A^2 = \frac{2}{a} \Rightarrow A = \sqrt{\frac{2}{a}}$$

Substitute "A" in wave function.

$$\psi(x) = A \sin kx$$

$$= \sqrt{\frac{2}{a}} \sin kx$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \left(\frac{n\pi}{a} \right) x$$

n= 1 (wave function for Ground state)

$$\psi_1(x) = \sqrt{\frac{2}{a}} \sin \left(\frac{\pi}{a} \right) x$$

n= 2 (Wave function for 1st excited state)

$$= \psi_2(x) = \sqrt{\frac{2}{a}} \sin \left(\frac{2\pi}{a} \right) x$$

n= 3 (Wave function for 2nd excited state)

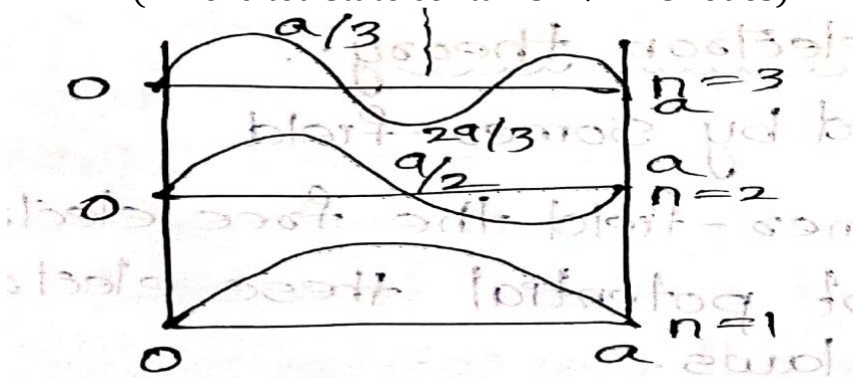


$$\psi_3(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi}{a}x\right)$$

If 'n' is the quantum energy level the wave function contains (n+1) nodes (minimum amplitude position).

Example: If n = 1 (Ground state energy level contains 1+1 nodes = 2 nodes)

if n = 2 (1st excited state contains 2+1 = 3 nodes)



UNIT II

1. Write any two assumptions given by Sommerfeld?

1. Valence electrons move freely in constant potential within the boundary of metal.
Hence, electrons are trapped in potential boxes.
2. The distribution of electrons in various energy levels occurs as per Pauli's exclusion principle.

Distribution of energy of electrons obeys fermi-dirac distribution.

2. Write any two Merits of classical free electron theory?

1. It explained electrical conductivity and thermal conductivity of metal.
2. It explained Wiedemann - Franz law.
3. It explained optical properties of metal



3. Write any two assumptions given by Sommerfield?

- 1) Valence electrons move free freely in constant potential with in the boundary of metal. Hence, electron trapped in potential box.
- 2) The distribution of electron in various energy level occurs as per Pauli's exclusion principle.

4. Write any two Merits of quantum free electron theory?

1. It explains electrical conductivity, thermal conductivity of metal.
2. It explains the variation of electrical conductivity & thermal with temperature.

5. Write any two De Merits of quantum free electron theory?

1. It fails to distinguish between conductors, insulators, semiconductor.
2. It fails to explain Hall effect phenomenon.

6. Write any two assumptions in classical free electron theory?

1. In Metal large no. of free electrons moves freely in all possible directions.
2. These electrons behave as gas molecules in container which obeys kinetic theory of gases.

7. Write any two Demerits of classical free electron theory?

1. It fails to explain electrical conductivity & thermal conductivity of semiconductors & Insulators.
2. It fails to explain the variation of electrical conductivity, thermal conductivity at low temperature.

8. Explain about the Insulators with examples?

The valence band & conduction bands are separated by a very large energy gap (E_g greater or equals to 3eV) At room temp conduction band is empty & valence band is full of electrons. Even at high temperature valence electrons unable to jump into conduction

Eg: Paper, Wood

9. Define Fermi energy level?



The Fermi level is the highest occupied energy level of electrons in a material at absolute zero temperature. It represents the energy level at which the probability of finding an electron is 50%.

10. Define Semiconductors with two examples?

The valence band and conduction bands are separated with a small energy band gap ($E_g = \text{lev}$). At low temperature (0K) free electrons are not available in conduction band hence they like an insulator. The electrical conductivity increases with increasing temperature thus the semiconductors behave as electrical conductors at high temperature. Example: Silicon $E_g = 1.1\text{eV}$
Germanium $E_g = 0.7\text{eV}$

Long questions

1. What are the assumptions, merits and Demerits of Quantum free electron theory

Quantum free electron theory:

Assumptions:

- It overcomes drawbacks of classical free electron theory.
- Sommerfeld explained them by using Fermi Dirac distribution instead of Maxwell-Boltzmann distribution.
 - 1) Valence electrons move freely in constant potential within the boundary of metal. Hence, electrons are trapped in potential box.
 - 2) The distribution of electrons in various energy levels occurs as per Pauli's exclusion principle.
- 3) Distribution of energy of electrons obeys fermi-dirac distribution.
- 4) The energy of electrons is quantized & discrete.

$$\text{i. e., } E_n = \frac{n^2 h^2}{8ma^2}$$

Merits:

- It explains electrical conductivity and thermal conductivity of metal.
- It explains the variation of electrical conductivity & thermal with temperature.
- It explains photoelectric effects, Compton effect & black body radiation.
- It explains specific heat of electron & susceptibility of para magnetic & ferromagnetic material with the temperature.

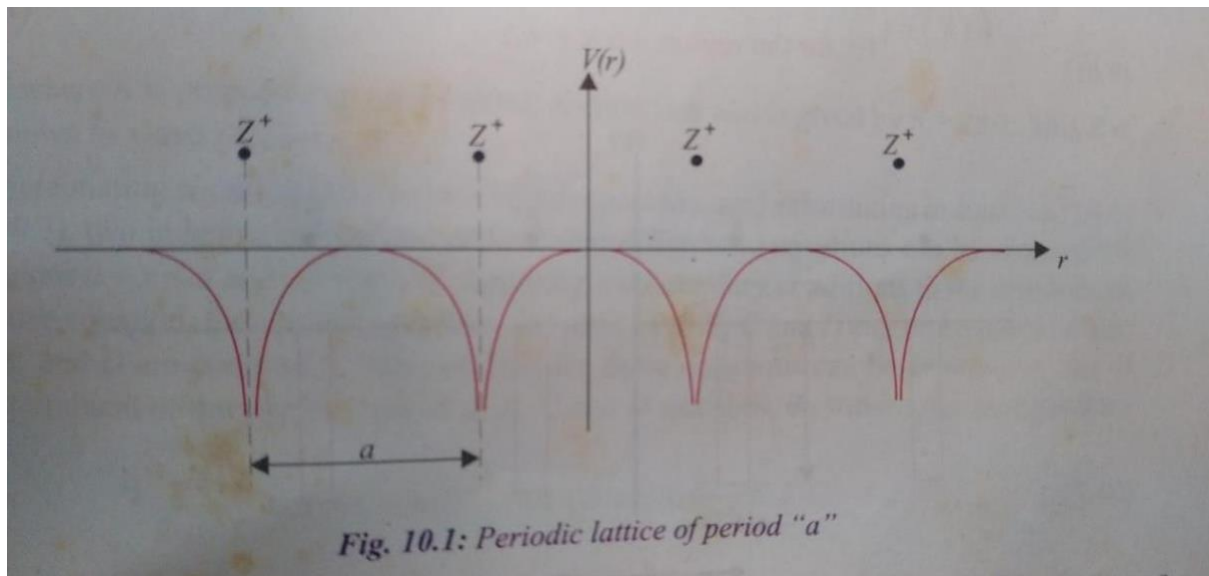
De-merits:



- It fails to distinguish between conductors, insulators, semiconductors.
- It fails to explain Hall effect phenomenon.
- According to quantum free electron theory only two electrons exist in fermi energy level, which are responsible for conductivity, but this is not true.

2. Using the Schrodinger time independent wave equation derive the expression for the Bloch equation?

- Let us consider a +ve ion core ('N' no of ions) with periodicity 'a' (distance between any two ions).
- Let's consider an electron moving freely between the positive ion core,



when the electron at the site of +ve ion the potential is minimum ($V=0$),
when electrons between the +ve ions the potential is maximum (V_0).

The motion of electrons is explained by schrodinger equation.

$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - V(x)]}{h^2} \psi(x) = 0 \quad \text{--- (i)}$$

$$\text{The potential for single ion } V(x) = V(x + a) \quad \text{--- (ii)}$$

Substitute (ii) in (i)

$$\frac{d^2\psi(x)}{dx^2} + 3\pi^2m \frac{[E - V(x + a)]}{h^2} \psi(x) = 0 \quad \text{--- (iii)}$$



- From the block the solution of equation (iii) can be written as multiplication of plane wave & periodicity.

$$\psi(x) = u_k(x)e^{ikx} \quad \text{-----} \quad (iv)$$

- If we consider 'N' no. of atom, then potential

$$V(x) = V(x + Na)$$

$$u_k(x) = u_k(x + Na) \text{-----} v$$

Then the solution from Bloch.

$$\psi(x + Na) = u_k(x + Na)e^{ik(x + Na)}$$

From equation number "v"

$$\psi(x + Na) = [u_k(x)]e^{iKx} \cdot e^{iKNa}$$

$$= u_k(x) \cdot e^{iKx} \cdot e^{iKNa} \Rightarrow \text{from equation (iv)}$$

$$\psi(x + Na) = \psi(x) \cdot e^{iKNa} \quad \text{-----} \quad (vi)$$

This is known as Bloch equation (or) theorem complex

Conjugate of equation (vi)

$$\psi^* \quad (x + Na) = \psi^* \cdot e^{-iKNa} \quad \text{-----} \quad (vii)$$

Multiply (vi) & (vii)

$$\psi(x + Na) \cdot \psi^*(x) = \psi(x)\psi^*(x)e^{-iKNa}$$

$$|\psi(x + Na)|^2 = |\psi(x)|^2$$

i.e., probability density for electrons is equal to entire crystal.

● 3. What are the assumptions and merits and demerits of classical free electron theory?

Assumptions:

- In Metal large no. of free electrons moves freely in all possible directions.
- These electrons behave as gas molecules in container which obeys kinetic theory of gases.
- In the absence of field, the energy associated with electron $E = \frac{3}{2}KT$

which is equal to kinetic energy

$$\frac{3}{2}KT = \frac{1}{2}mv_{th}^2$$

V_{th} = Thermal velocity.

- Metal consists of positive ion core in fixed position and electron moves freely collides with positive ion core (or) free electron (or) with the metal boundary.
- These collisions are perfectly elastic collisions. Hence, conductivity is due to free electrons.
- The electron velocity obeys Maxwell-Distribution velocity.
- The potential energy of electron is always constant value.



- If electric field is applied the electron gets accelerated, the velocity of electron opposite direction of electric field this velocity is called drift velocity (V_d).

Merits:

- It explained Ohm's law.
- It explained electrical conductivity, thermal conductivity of metal.
- It explained Wiedemann - Franz law.
- It explained optical properties of metal.

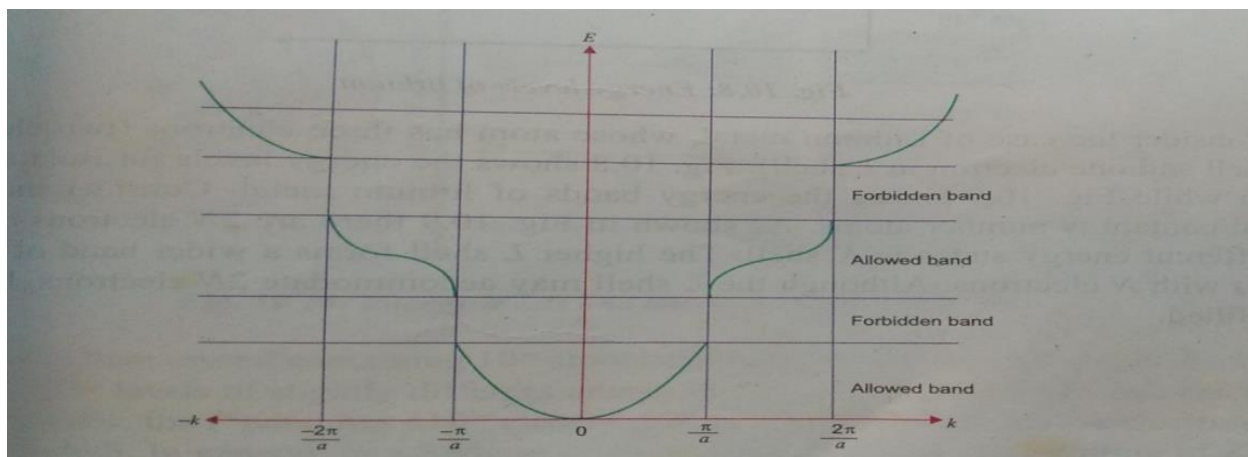
De-Merits:

- It fails to explain electrical conductivity & thermal conductivity of semiconductors & Insulators.
- It fails to explain the variation of electrical conductivity, thermal conductivity at low temperature.
- It fails to explain photoelectric effect, black body radiation and specific heat of electron.
- It fails to explain the susceptibility of paramagnetic material and Ferromagnetic material based on temperature.

4. Draw the E-K Diagram and Define the Brillouin Zone and forbidden zone?

E – k diagram by using $P\sin\alpha a + \cos\alpha a = \cos ka$ we can plot the curve (graph) energy as a function of wave vector

In above equation right hand side term ($\cos ka = \pm 1$) for the values of $k = \pm h\pi / a$ Hence, discontinuity occurs in energy spectrum consist of allowed region and forbidden region.



If $n = 1$, $k = \pm h\pi / a$ then allowed region extended $-\pi / a$ to π / a called first Brillouin zone.

If $n = 2$, $k = \pm 2\pi / a$ i. e., $k = -2\pi / a, -\pi / a, 0$ and $\pi / a, 2\pi / a$ called second Brillouin zone.



In an **E-k diagram**, the Brillouin Zone defines the range of wave vectors (k) within which the electron energy bands are plotted.

The x-axis represents the wave vector (k), related to the momentum of electrons.

The first Brillouin Zone typically spans from $-\pi/a$ to $+\pi/a$ (in 1D), where a is the lattice constant.

The edges of the Brillouin Zone (e.g., $\pm\pi/a$) are high-symmetry points where energy band gaps may occur due to Bragg reflection of electron waves.

In the diagram:

Brillouin Zones appear as regions between these symmetry points. Energy bands are typically shown folding back at the Brillouin zone boundaries.

Forbidden Energy Band Gap in E-k Diagram:

In an E-k diagram, the forbidden energy band gap is a vertical energy range between two bands where no electronic states exist.

It appears between the top of the valence band and the bottom of the conduction band.

The y-axis (energy E) shows this gap clearly as a region where there are no allowed energy levels

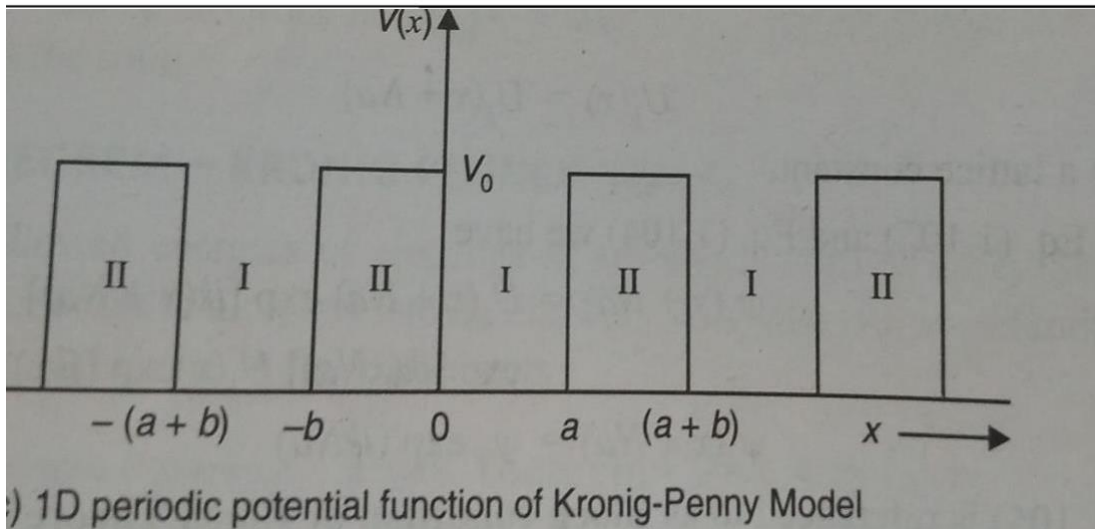
for electrons.

The gap often occurs at the Brillouin zone boundary (e.g., at $k = \pm\pi/a$) due to the periodic potential of the crystal.

5.Explain the Kronig-Penny model for the electron in periodic potential?

They proposed periodic potential.

They approximated periodicity to a 1D rectangular potential with the periodicity $(a+b)$.



The motion of electron is given by schrodinger wave equation.



$$\frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - V(x)]}{h^2} \psi(x) = 0 \quad \text{--- (i)}$$

Potential:

In region-I $V(x) = 0 \quad 0 < x < a$

region-II $V(x) = V_0 \quad -b < x < 0$

By applying Schrodinger equation in region-I.

$$\begin{aligned} \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - 0]}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2mE}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} + \alpha^2 \psi(x) &= 0 \quad \text{--- (ii)} \end{aligned}$$

Apply Schrodinger wave equation region-II.

$$\begin{aligned} V(x) &= V_0, \quad -b < x < 0 \\ \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[E - V_0]}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2m[V_0 - E]}{h^2} \psi(x) &= 0 \\ \frac{d^2\psi(x)}{dx^2} - \beta^2 \psi(x) &= 0 \quad \text{--- (iii)} \\ \beta^2 &= \frac{8\pi^2m(V_0 - E)}{h^2} \end{aligned}$$

The solution of equation (ii) & equation (iii) from Bloch can be written as

$$\psi(x) = u_k(x)e^{iKx} \quad \text{--- (iv)}$$

after applying boundary conditions by mathematical expression written as

$$\frac{2\alpha\beta}{\beta^2 - \alpha^2} \sin\alpha a \sinh(\beta b) + \cos\alpha a \cosh(\beta b) = \cos k(a + b) \quad \text{--- (v)}$$

While increasing potential of the barrier then width of potential barrier decreases.

$$V_0 \Rightarrow \infty \Rightarrow b = 0$$

$$\text{Then } \sinh(\beta b) \rightarrow \beta b$$

$$\cosh(\beta b) \rightarrow 1$$

Substitute in equation (v)

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sin\alpha a (\beta b) + \cos\alpha a (1) = \cos k(x) \quad \text{--- (vi)}$$

We know that

$$\begin{aligned} \beta &= \frac{8\pi^2m[V_0 - E]}{h^2}; \alpha^2 = \frac{3\pi^2mE}{h^2} \\ \beta^2 - \alpha^2 &= \frac{8\pi^2m(V_0 - E)}{h^2} - \frac{8\pi^2mE}{h^2} \\ &= \frac{8\pi^2m}{h^2} (V_0 - E - E) \\ &= \frac{8\pi^2m}{h^2} (V_0 - 2E) \end{aligned}$$

$$V_0 \gg E$$



$$\beta^2 - \alpha^2 = \frac{8\pi^2 m V_0}{h^2} \quad \text{substitute in (vi)}$$

$$\frac{8\pi^2 m V_0}{h^2 \cdot \alpha \beta} \beta b \sin \alpha a + \cos \alpha a = \cos k a$$

$$\frac{h^2 \cdot \alpha}{4\pi^2 m V_0 b a} b \sin \alpha a + \cos \alpha a = \cos k a$$

$$\frac{h^2 \alpha a}{4\pi^2 m V_0 b a} \sin \alpha a + \cos \alpha a = \cos k a$$

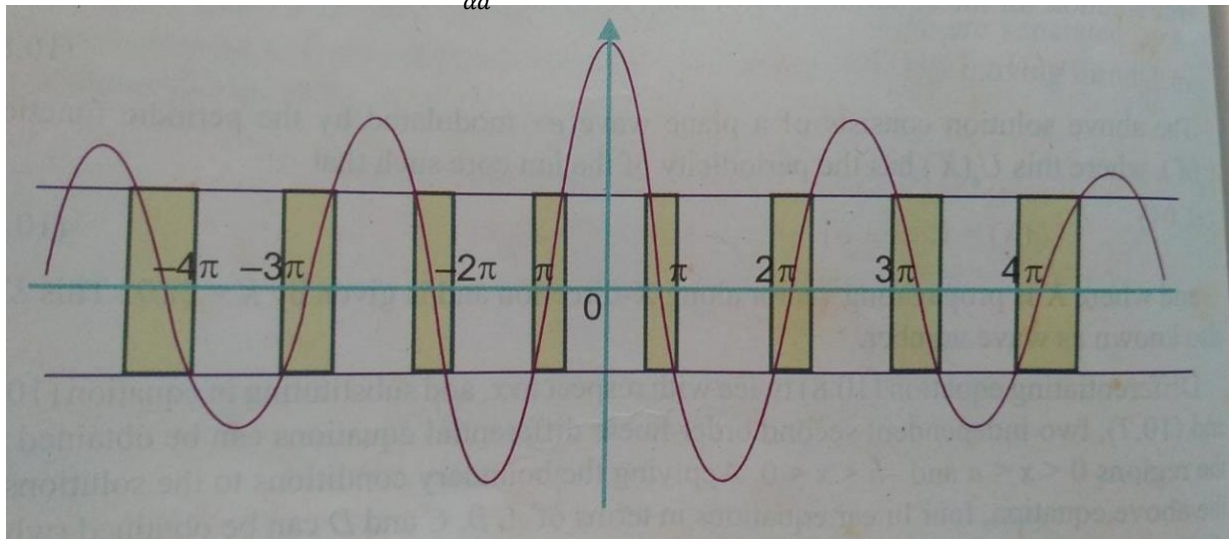
$$\left[\frac{h^2 \alpha a}{4\pi^2 m V_0 b a} \right] \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos k a$$

$P \rightarrow$ Scattering power (or) Barrier strength.

$$\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a = \cos k a \quad \text{--- (vii)}$$

➤ It will give attractions between electron and the +ve electron core.

By plotting graph between $\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a$ and αa then graph will be



➤ From the graph electron characterized in an energy level corresponding to particular value of αa . This leads the existence of allowed region and forbidden region decreases.

➤ If P is very large ($P \rightarrow \infty$) P is barrier strength.

from equation (vii)

$$\frac{\sin \alpha a}{\alpha a} = 0$$

$$\sin \alpha a = 0$$

$$\alpha a = \sin^{-1}(0)$$

$$\alpha a = \sin^{-1}(\sin n\pi)$$

$$\alpha a = n\pi$$

$$\alpha = \frac{n\pi}{a}$$

$$\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a = \cos k a$$

$$[\because P \rightarrow \infty]$$

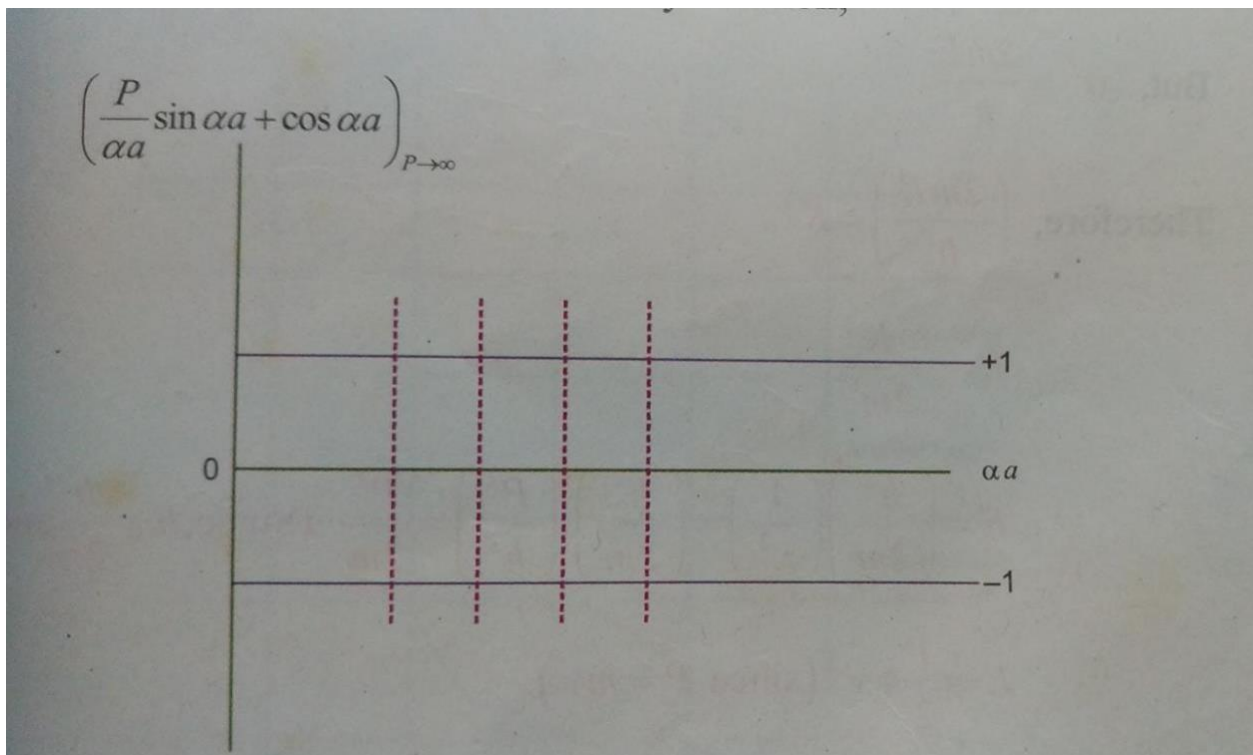


$$\Rightarrow \alpha^2 = \frac{n^2 \pi^2}{8\pi^2 m E} = \frac{a^2}{h^2} = \frac{n^2 h^2}{8ma^2}$$

$$E = \frac{n^2 h^2}{8ma^2}$$

i.e., energy is quantized for when barrier strength is more.

➤ with increase in barrier strength (P) width of allowed region decreases.



➤ P is very small $P \rightarrow 0$

$$\frac{P \sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

$$\cos \alpha a = \cos ka$$

$$\alpha a = ka \quad [\because P = 0]$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

$$\frac{8\pi^2 m E}{h^2} = \frac{4\pi^2}{\lambda^2}$$

$$\frac{2mE}{h^2} = \frac{1}{\lambda^2}$$

$$\frac{h^2}{h^2} = \frac{1}{\lambda^2}$$

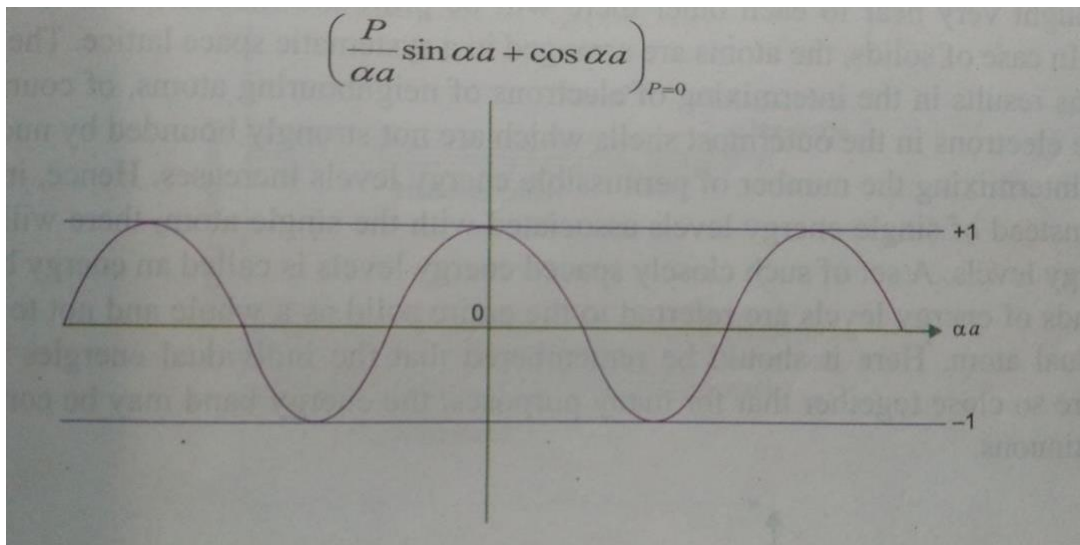
$$[\because k = \frac{2\pi}{\lambda}]$$

$$[\because \lambda = \frac{h}{mv}]$$

$$\frac{2mE}{h^2} = \frac{m^2 u^2}{h^2}$$

$$E = \frac{1}{2} m v^2$$

i.e., the total energy of the electron is from the kinetic energy, potential energy is zero then the particle is free particle then the energy spectrum.



i.e., electrons are allowed in all energy band this support classical free electron theory.

6. Give a brief explanation about conductors, semiconductors and insulators?

Conductors:

The valence band and conduction bands are overlapped with each other so energy gap (E_g) is zero.

At room temperature free electrons already exist in conduction band. Hence solids (metals) are electrical as well as thermal conductors.

· The electrical resistivity increases with increasing temperature in conductors.

Semiconductors:

The valence band and conduction bands are separated with a small energy band gap ($E_g = 1\text{eV}$).

At low temperature (0K) free electrons are not available in conduction band hence they like an insulator.

The electrical conductivity increases with increasing temperature thus the semiconductors behave as electrical conductors at high temperature.

Example: Silicon $E_g = 1.1\text{eV}$

Germanium $E_g = 0.7\text{eV}$

Insulators:

The valence band & conduction bands are separated by a very large energy gap (E_g greater or equals to 3eV)

At room temp conduction band is empty & valence band is full of electrons. Even at high temperature valence electrons unable to jump into conduction band.

UNIT III

Short Question & Answers:



1. Give the Introduction of Quantum Computing?

A normal (classical) computer stores and processes information using bits, which can be either 0 or 1. Everything your phone, laptop, or any computer does comes from billions of bits flipping between 0 and 1. A quantum computer works differently—it uses quantum bits, or qubits, which can be 0, 1 or both at the same time.

2. Write the notation of Bra-Ket?

Bra notation is written as $\langle\psi|$ (pronounced "bra psi").

A bra is the complex conjugate transpose of a ket. Mathematically, it acts as a row vector that can be paired with a ket to compute an inner product.

In mathematical terms, if $|\psi\rangle$ is a ket, its corresponding bra is $\langle\psi|$.

The bra is a function that maps a ket (vector) to a complex number and is used for measuring overlaps between quantum states.

3. Write two advantages of Quantum computing?

Faster Processing: Quantum computers can solve certain complex problems much faster than classical computers, such as factoring large numbers or simulating quantum systems.

Parallelism: Quantum bits (qubits) can exist in multiple states simultaneously (superposition), allowing quantum computers to perform many calculations at once.

4. Write the differences of Classical & quantum bits?

Classical Bits: A classical bit is the basic unit of information in traditional computers. It can only be in one of two states: 0 or 1, like a basic light switch being either off or on.



Computers use billions of these bits to store and process all their data and instructions. Classical bits follow the rules of ordinary physics (classical physics) and are manipulated by logic gates like AND, OR, and NOT.

Quantum Bits (Qubits) A quantum bit, or qubit, is the fundamental unit in quantum computers. Qubits can be 0, 1, or both at the same time (through a property called superposition).

Because of this, qubits can process lots of information at once, making quantum computers very powerful for certain tasks.

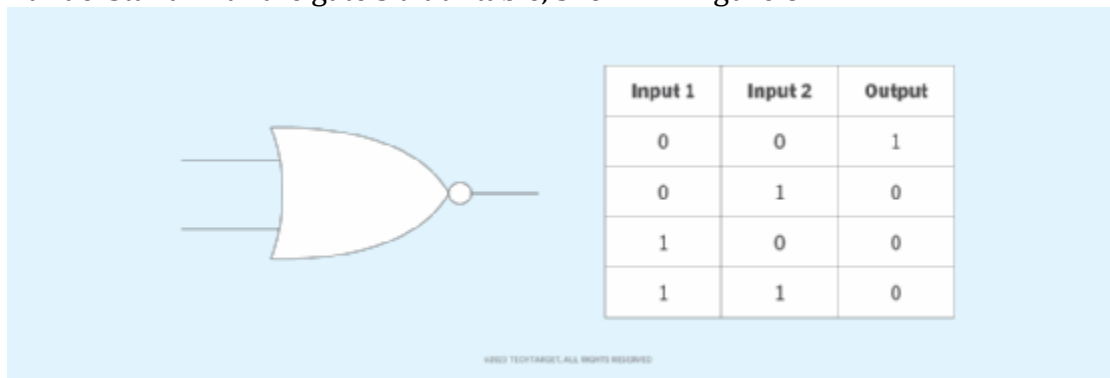
Qubits also show entanglement, where the state of one qubit can depend on another, even at a distance.

They operate under the rules of quantum mechanics, not classical physics, and are manipulated by quantum logic gates.

5. Explain classical logic gates NOR with truth tables.

NOR gate

The NOR, or NOT OR, gate is a combination OR gate followed by an inverter. Its output is true or high if both inputs are false or low (logic 0). Otherwise, the output is false. This logic is easy to understand with the gate's truth table, shown in Figure 6.



6. Explain classical logic gates NAND with truth tables.

NAND gate

The negated AND, or NAND, gate operates as an AND gate followed by a NOT gate. Its symbol is an AND gate with the circle of a NOT gate at the output. The NAND gate produces a false or low output if both inputs are true. Otherwise, the output is true. Another way to visualize it is that a NAND gate inverts the output of an AND gate. It acts in the manner of the logical operator *and* followed by negation. Figure 5 provides a truth table showing the logic of a NAND gate for two types of inputs.



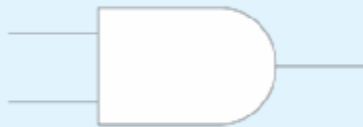
Input 1	Input 2	Output
0	0	1
0	1	1
1	0	1
1	1	0

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7.Explain classical logic gates AND with truth tables

AND gate

The AND gate produces an output of true only when both inputs are true. Otherwise, the output is false. In other words, the output is 1 only when both inputs are 1, and 0 if even one of the inputs is 0. The gate is named so because, if 0 is false and 1 is true, the gate acts in the same way as the logical *and* operator. Figure 1 provides a [truth table](#) showing the circuit symbol and logic combinations for an AND gate. In the symbol, the input terminals are on the left, and the output terminal is on the right.



Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

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8.Explain classical logic gates OR with truth tables

OR gate

In the OR gate, which is among the most widely used logic circuits in digital devices and [components](#), the output is true if one or both inputs are true. If both inputs are false, the output is false. For the output to be 1, at least one input must be 1. The gate gets its name from behaving like the logical inclusive *or*. Figure 2 shows a truth table for the OR gate.



Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	1

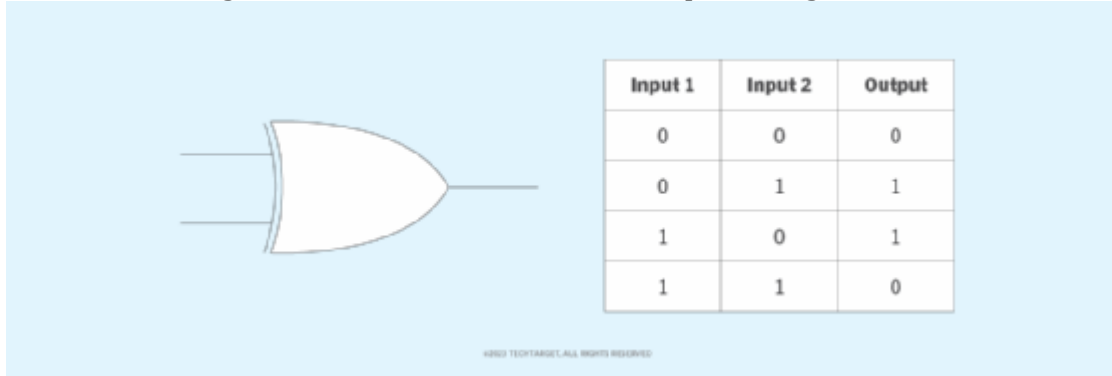
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9.Explain classical logic gates XOR with truth tables



XOR gate

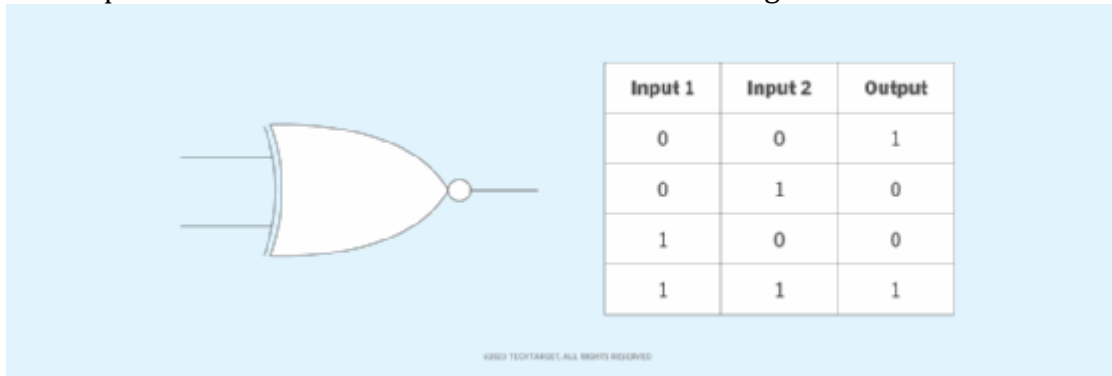
The XOR, or exclusive OR, gate acts in the same way as the logical *either/or*. The output is true if either input is true, but not both. The output is false if both inputs are false or both inputs are true. Similarly, the output is 1 (true or high) if the inputs are different, but 0 (false or low) if the inputs are the same. Figure shows a truth table for a 2-input XOR gate.



10.Explain classical logic gates XNOR with truth tables

XNOR gate

The XNOR, or exclusive NOR, gate is a combination of an XOR gate followed by an inverter. Its output is true if the inputs are the same and false if the inputs are different. It produces a true output, regardless of whether the inputs are true or false. However, they must both be the same for the output to be true. The XNOR truth table looks like Figure 7.



Long Questions

1. Give brief explanation about Superposition and Entanglement?

Superposition

Superposition is a unique property of quantum systems where a quantum bit (qubit) can exist in multiple states simultaneously.

Unlike a classical bit, which must be either 0 or 1, a qubit can be in a state that is both 0 and 1 at the same time until it's measured.

Think of it like a coin spinning in the air—until you look (measure), it's both heads and tails at once.

Superposition is what allows quantum computers to process many possibilities all at the same time, vastly increasing their potential computational power for certain tasks.



When you measure (read) a qubit, the superposition collapses, and you find it in one of its possible states.

Entanglement

Entanglement is another fundamental concept in quantum mechanics.

It's a special relationship between two or more qubits, where the state of one qubit instantly affects the state of the other, no matter how far apart they are.

If two qubits are entangled, measuring one immediately gives you information about the other,

even if they're separated by huge distances. I

n quantum computing, entanglement allows for linked qubits that work together in ways classical

bits can't.

This property is crucial for the power and complexity of quantum algorithms and for secure quantum communication.



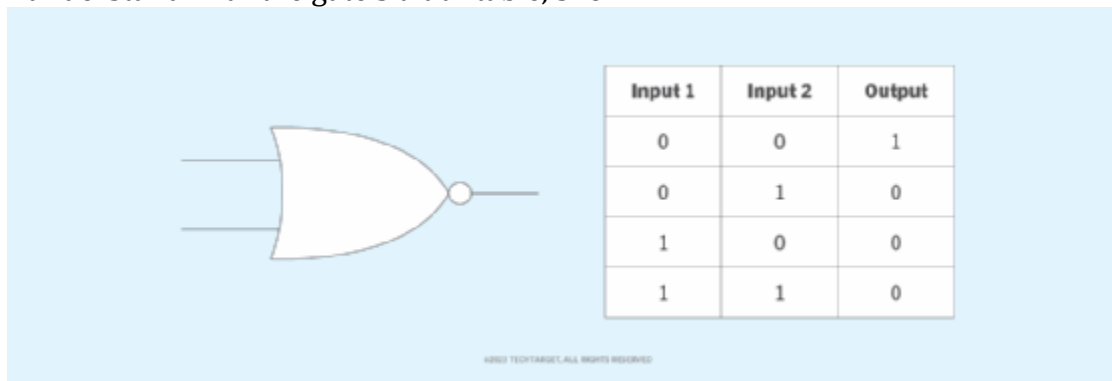
This concept is often illustrated by the famous "Schrödinger's cat" thought experiment and the double-slit experiment.

In the latter, a single particle, like a photon, goes through two slits simultaneously and creates an interference pattern—showing that it's a superposition of going through both slits at once.

2. Give the explanation of the Universal gates?

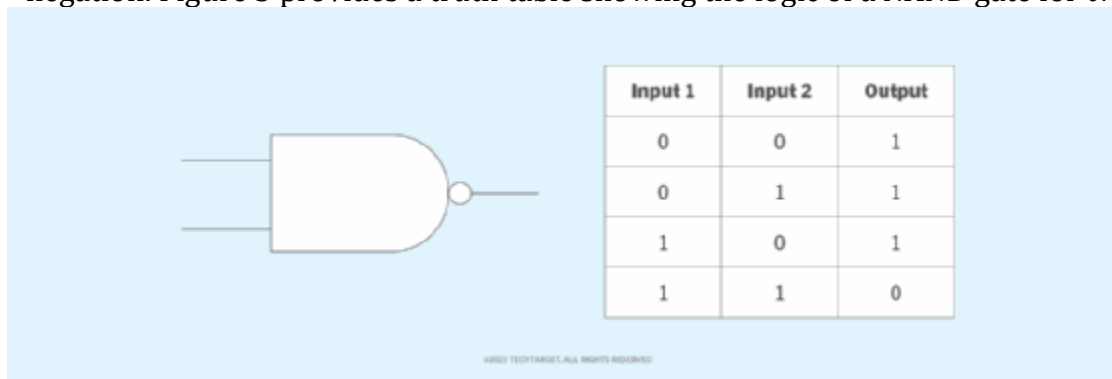
NOR gate

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NAND gate

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3. Give the explanation of the Basic classical gates ?

AND gate

The AND gate produces an output of true only when both inputs are true. Otherwise, the output is false. In other words, the output is 1 only when both inputs are 1, and 0 if even one of the inputs is 0. The gate is named so because, if 0 is false and 1 is true, the gate acts in the same way as the logical *and* operator. Figure 1 provides a [truth table](#) showing the circuit symbol and logic combinations for an AND gate. In the symbol, the input terminals are on the left, and the output



terminal is on the right.



Input 1	Input 2	Output
0	0	0
0	1	0
1	0	0
1	1	1

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OR gate

In the OR gate, which is among the most widely used logic circuits in digital devices and [components](#), the output is true if one or both inputs are true. If both inputs are false, the output is false. For the output to be 1, at least one input must be 1. The gate gets its name from behaving like the logical inclusive *or*. Figure 2 shows a truth table for the OR gate.



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0	1	1
1	0	1
1	1	1

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XOR gate

The XOR, or exclusive OR, gate acts in the same way as the logical *either/or*. The output is true if either input is true, but not both. The output is false if both inputs are false or both inputs are true. Similarly, the output is 1 (true or high) if the inputs are different, but 0 (false or low) if the inputs are the same. Figure shows a truth table for a 2-input XOR gate.



Input 1	Input 2	Output
0	0	0
0	1	1
1	0	1
1	1	0

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XNOR gate

The XNOR, or exclusive NOR, gate is a combination of an XOR gate followed by an inverter. Its output is true if the inputs are the same and false if the inputs are different. It produces a true output, regardless of whether the inputs are true or false. However, they must both be the same for the output to be true. The XNOR truth table looks like Figure 7.



Input 1	Input 2	Output
0	0	1
0	1	0
1	0	0
1	1	1

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4. Write the matrix representations of Hadamard (H) gate

Hadamard Gate (H-gate) – It is a single qubit gate and is also gate to superposition. H gate acts on single qubit input and produce superposition state output. Matrix form of H gate and its symbol is

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



Action of H gate on inputs:

Let us find out what happens when Hadamard gate operates on a qubit that is in the $|0\rangle$ state

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$
$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Let us find out what happens when Hadamard gate operates on a qubit that is in the $|1\rangle$ state.

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$
$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$



If a superposed qubit goes through H gate, the result will be

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix} = \left(\frac{\alpha + \beta}{\sqrt{2}}\right)|0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}}\right)|1\rangle$$
$$H|\psi\rangle = \alpha\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

Gate representation is

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{H} \longrightarrow \alpha\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

The truth table is as follows

Input	Output
$ 0\rangle$	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$
$ 1\rangle$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$
$\alpha 0\rangle + \beta 1\rangle$	$\alpha\left(\frac{ 0\rangle + 1\rangle}{\sqrt{2}}\right) + \beta\left(\frac{ 0\rangle - 1\rangle}{\sqrt{2}}\right)$

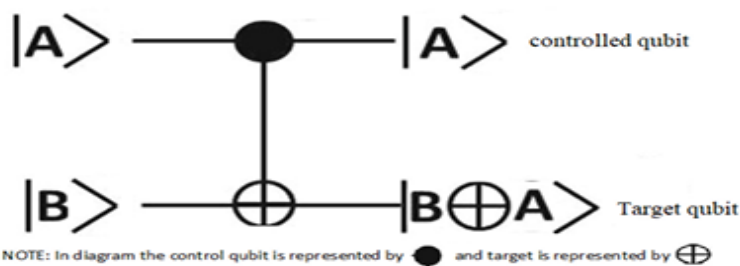
5.Explain the working of the CNOT (Controlled-NOT) gate with its circuit and matrix form.



1. Controlled Gate (CNOT)

The CNOT gate is a two-qubit operation, where the first qubit is referred as the control qubit $|A\rangle$ and the second qubit as the target qubit $|B\rangle$. If the control qubit is $|1\rangle$ then it will flip the target qubit state from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$. When the control qubit is in state $|0\rangle$ then the target qubit remains unchanged.

The symbolic representation is as follows. The upper line represents control qubit and bottom line represents target qubit.



In the combined qubit, first term is control qubit and the second term is target qubit. For ex, in $|AB\rangle$, A is control qubit and B is target qubit

Matrix form of **CNOT** Gate is given by

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The Transformation could be expressed as $|A, B\rangle \rightarrow |A, B \oplus A\rangle$

Action of the gate: Consider the operations of CNOT gate on the four inputs $|00\rangle, |01\rangle, |10\rangle$ and $|11\rangle$.

- i) Operation of CNOT Gate for input $|00\rangle$: $|00\rangle \rightarrow |00\rangle$

Here, control qubit is $|0\rangle$. Hence no change in the state of Target qubit $|0\rangle$.



- ii) Operation of CNOT Gate for input $|01\rangle$: $|01\rangle \rightarrow |01\rangle$
Here, control qubit is $|0\rangle$. Hence no change in the state of Target qubit $|1\rangle$
- iii) Operation of CNOT Gate for input $|10\rangle$: $|10\rangle \rightarrow |11\rangle$
Here, control qubit is $|1\rangle$. Hence the state of Target qubit flips from $|0\rangle$ to $|1\rangle$.
- iv) Operation of CNOT Gate for input $|11\rangle$: $|11\rangle \rightarrow |10\rangle$
Here control qubit is $|1\rangle$. Hence the state of Target qubit flips from $|1\rangle$ to $|0\rangle$.

Input		Output	
$ A\rangle$	$ B\rangle$	$ A\rangle$	$ B \oplus A\rangle$
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$

The Truth Table of operation of CNOT gate is as follows.

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

6. Write the matrix representations of Pauli-X, Pauli-Y, and Pauli-Z gates



Single qubit gates

1. X – Gate

This is also called as Pauli X – gate. It is given by

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

When X operates on $|0\rangle$ and $|1\rangle$ the output will be inverted (ie, $|0\rangle$ becomes $|1\rangle$ and $|1\rangle$ becomes $|0\rangle$)

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Since X inverts each input it is also called as *bit-flip* gate. If a superposed qubit goes through X gate, the result will be

$$X|\Psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$$

So,

$$X|\Psi\rangle = \alpha|1\rangle + \beta|0\rangle$$

Symbolically these gates are represented as follows



2. Y – Gate

This is also called as Pauli Y – gate. It is given by

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

When Y operates on $|0\rangle$ and $|1\rangle$

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \times 1 + (-i) \times 0 \\ i \times 1 + 0 \times 0 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$Y|1\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \times 0 + (-i) \times 1 \\ i \times 0 + 0 \times 1 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -i|0\rangle$$

So,

$$Y|0\rangle = i|1\rangle \text{ and } Y|1\rangle = -i|0\rangle$$



If a superposed qubit goes through Y gate, the result will be

$$Y|\Psi\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \times \alpha + (-i) \times \beta \\ i \times \alpha + 0 \times \beta \end{bmatrix} = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix} = -i\beta|0\rangle + i\alpha|1\rangle$$

So,

$$Y|\Psi\rangle = i\alpha|1\rangle - i\beta|0\rangle$$

Symbolically these gates are represented as follows



3. Z – Gate

This is also called as Pauli Z – gate. It is given by

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

When Z operates on $|0\rangle$ and $|1\rangle$ the phase will change. Hence this is also called as *phase-flip* gate

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 0 \times 0 \\ 0 \times 1 + (-1) \times 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 0 \times 1 \\ 0 \times 0 + (-1) \times 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

So,

$$Z|0\rangle = |0\rangle \text{ and } Z|1\rangle = -|1\rangle$$

If a superposed qubit goes through Z gate, the result will be

$$Z|\Psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 \times \alpha + 0 \times \beta \\ 0 \times \alpha + (-1) \times \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

So,

$$Z|\Psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

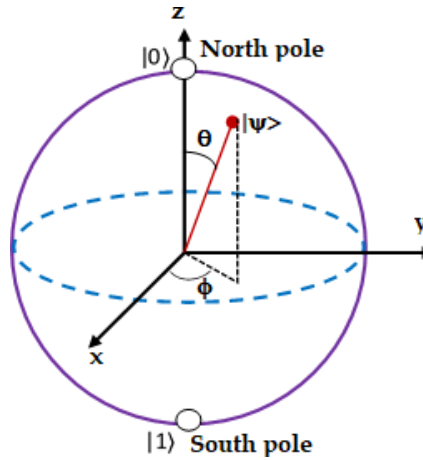
Symbolically these gates are represented as follows



7.Explain the Bloch sphere representation of a qubit with a neat diagram.

Bloch sphere

Bloch sphere is an imaginary sphere which is used to represent pure single-qubit states as a point on its surface. It has unit radius. Its North Pole and South Pole are selected to represent the basis states namely $|0\rangle$ and $|1\rangle$. North Pole represents $|0\rangle$ (say spin up $\uparrow$) and South Pole represents $|1\rangle$ (say spin down $\downarrow$). All other points on the sphere represent superposed states (ie, state space). Bloch sphere allows the state of a qubit to be represented in spherical coordinates (ie, r , θ and ϕ). It is as follows



The state qubit $|\psi\rangle$ on the Bloch sphere makes an angle θ with z-axis and its projection (azimuth) makes angle ϕ with x-axis as shown. It is clear from the fig that $0 < \theta < \pi$ and $0 < \phi < 2\pi$. $|\psi\rangle$ is represented as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

It can be proved that

$$|\Psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \quad \text{--- (1)}$$

Using this equation we can represent $|\psi\rangle$ for different θ and ϕ as follows

Case-1: let $\theta = 0$ and $\phi = 0$, then eq (1) becomes

$$\begin{aligned} |\Psi\rangle &= \cos 0 |0\rangle + e^{i0} \sin 0 |1\rangle = |0\rangle + 0 \\ \therefore |\Psi\rangle &= |0\rangle \end{aligned}$$

This is the **north pole**, representing the base state $|0\rangle$.

Case-2: let $\theta = \pi$ and $\phi = 0$, then eq (1) becomes

$$\begin{aligned} |\Psi\rangle &= \cos \frac{\pi}{2} |0\rangle + e^{i0} \sin \frac{\pi}{2} |1\rangle = 0 + |1\rangle \\ \therefore |\Psi\rangle &= |1\rangle \end{aligned}$$

This is the **south pole**, representing the base state $|1\rangle$.

Case-3: let $\theta = \pi/2$ and $\phi = 0$, then eq (1) becomes

$$\begin{aligned} |\Psi\rangle &= \cos \frac{\pi}{4} |0\rangle + e^{i0} \sin \frac{\pi}{4} |1\rangle \\ |\Psi\rangle &= \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \\ |\Psi\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \end{aligned}$$

This point lies on the **positive x-axis**, representing the superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.



Case-4: let $\theta = \pi/2$ and $\phi = \pi$, then eq (1) becomes

$$\begin{aligned} |\Psi\rangle &= \cos \frac{\pi}{4} |0\rangle + e^{i\pi} \sin \frac{\pi}{4} |1\rangle \\ |\Psi\rangle &= \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \\ |\Psi\rangle &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned}$$

This point lies on the **negative x-axis**, representing the superposition $|- \rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

In the above discussion we have represented only single qubit state. Bloch sphere is a nice visualization of single qubit states.

The Bloch sphere is a vital visual model that represents a single qubit's state as a point on a sphere, where its position defines the qubit's probability and phase.

Unit IV

Short Question

1. What is the main difference between classical circuits and quantum circuits

<u>Classical Circuits</u>	<u>Quantum Circuits</u>
① Basic unit of Classical is Bit (0 or 1)	① Basic unit of Quantum is Qubit (Superposition of 0 and 1)
② State representation of Classical single definite value (0 or 1)	② State representation of Quantum is vector in Hilbert space Ψ : Ex: $\alpha 0\rangle + \beta 1\rangle$
③ Classical operate based on Boolean algebra and deterministic logic gates AND, OR etc.	③ Quantum Linear algebra matrices, tensors, vectors, Unitary operators.
④ Classical most gates are irreversible	④ Quantum All gates must be reversible.

2. What are the basic notations used in quantum circuits, such as $|0\rangle$, $|1\rangle$, and gate symbols?



Basic circuit notation :-

A Quantum circuit is a diagrammatic representation of how qubits are processed in a quantum computation.

The basic notations used in circuits so that operations can be easily understood.

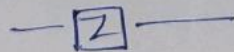
→ X gate : Flips the qubit



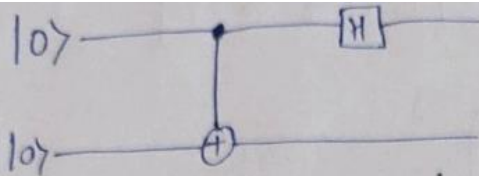
→ H gate : Creates superposition,



→ Z and Y gates: Apply phase and rotation changes.



3. Draw the circuit for non-entanglement



A CNOT gate is applied to the $|00\rangle$ state followed by a Hadamard gate on the first qubit.

$$\psi_i = |00\rangle$$

$$\psi_f = (H \otimes I) \text{ CNOT } |00\rangle$$



4. Explain the Introduction to quantum circuits

Introduction to Quantum Circuits:

A Quantum Circuit is a sequence of Quantum gates applied to a set of Qubits. Qubits can exist in a superposition of both states simultaneously.

This property along with Entanglement and Interference, allows Quantum Circuits to perform complex computations more efficiently than classical circuits for certain problems.

5. Draw the circuit diagram to create a superposition using H and NOP

Circuit diagram.

$|0\rangle$ — $[H]$ —

$|1\rangle$ —

$\psi_f = (H \otimes I) (|0\rangle \otimes |1\rangle)$

$\psi_f = (H \otimes I) |01\rangle$

$|0\rangle$ — $[H \otimes I]$ — ψ_f

6. Draw the circuit diagram to create a Bell pair using H and CNOT gates.

$|0\rangle$ — $[H]$ —

$|0\rangle$ —

$\psi_i = |00\rangle$

A Hadamard gate is applied to the first qubit, followed by a CNOT gate.

$\psi_f = \text{CNOT} (H \otimes I) |00\rangle$



7. Draw the circuit diagram to create a superposition using H and H?

$$\begin{array}{l} |0\rangle \text{ --- [H] ---} \\ |1\rangle \text{ --- [H] ---} \end{array}$$

8. Draw the Parallel quantum circuit diagram by using X gate and Y gate

$$\begin{array}{l} |0\rangle \text{ --- [X] ---} \\ |1\rangle \text{ --- [Y] ---} \end{array}$$

9. Write any four points about quantum circuits ?

- ① Basic unit of Quantum is Qubit (Superposition of 0 and 1)
- ② State representation of Quantum is vector in Hilbert Space $\mathbb{C}^n$:
eg: $\alpha|0\rangle + \beta|1\rangle$
- ③ Quantum Linear algebra matrices, tensors, vectors, Unitary operators.
- ④ Quantum All gates must be reversible.

10. Write any four points about classical circuits ?



- ① Basic unit of Classical is Bit (0 or 1)
- ② State representation of Classical single definite value (0 or 1)
- ③ Classical operate based on Boolean algebra and deterministic logic gates AND, OR etc.
- ④ Classical most gates are irreversible

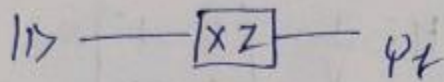
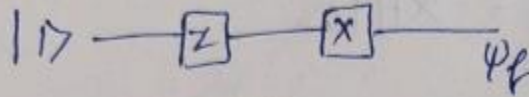
Long Question & Answers:

1. Find the final qubit state for the circuit $[Z] \rightarrow [X]$ Using inputs $|0\rangle$ and $|1\rangle$



Apply Z gate followed by X gate on $|1\rangle$

Consider the circuit



$$\psi_i = |1\rangle, \quad \psi_f = XZ|1\rangle$$

Matrices are

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

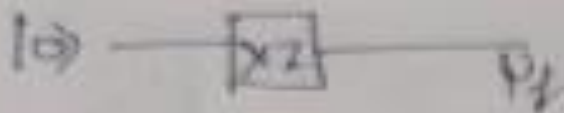
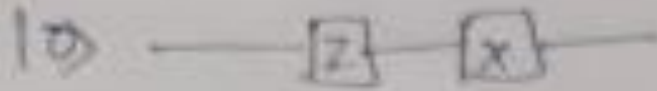
$$XZ|1\rangle = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -|0\rangle$$

$$\psi_f = XZ|1\rangle$$

$$\psi_f = -|0\rangle$$



Z-X on $|0\rangle$



$$\psi_i = |0\rangle \quad \psi_f = XZ|0\rangle$$

$$XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$XZ|0\rangle = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$\psi_f = XZ|0\rangle = |1\rangle$$

2. Find the final qubit state for the circuit $\text{---}[H]\text{---}[X]\text{---}$ Using inputs $|0\rangle$ and $|1\rangle$



Apply Hadamard (H) gate followed by X gate on $|0\rangle$

Circuit diagram.

$$|0\rangle \longrightarrow [H] \longrightarrow [X] \longrightarrow \psi_f$$

$$\psi_i \longrightarrow [XH] \longrightarrow \psi_f$$

$$\psi_i = |0\rangle, \quad \psi_f = XH|0\rangle$$

Matrices $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$XH = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$XH|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right]$$

$$= \frac{1}{\sqrt{2}} \left[|0\rangle + |1\rangle \right]$$

$$\psi_f = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

H creates a superposition and X flips the components but keeps the superposition same.



$$\underline{H-X \text{ on } |1\rangle}$$

$$|1\rangle \rightarrow \boxed{H} \rightarrow \boxed{X}$$

$$\psi_i |1\rangle \Rightarrow \psi_f = XH|1\rangle$$

$$XH|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$XH|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left[-\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right]$$

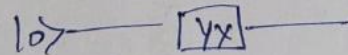
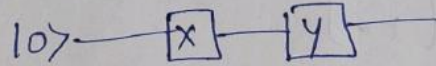
$$= \frac{1}{\sqrt{2}} \left[-|0\rangle + |1\rangle \right]$$



3. Find the final qubit state for the circuit —[X]—[Y]— Using inputs $|0\rangle$ and $|1\rangle$

Applying X gate followed by Y gate on $|0\rangle$

Consider the circuit



$$\psi_i = |0\rangle, \quad \psi_f = YX|0\rangle$$

matrices of the Pauli gate are

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$YX = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

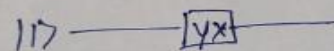
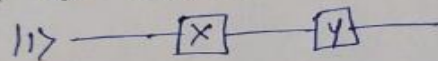
$$YX|0\rangle = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -i|0\rangle$$

$$\psi_f = YX|0\rangle$$

$$\psi_f = -i|0\rangle$$

Apply X gate followed by Y gate on $|1\rangle$

Consider the circuit



$$\psi_i = |1\rangle, \quad \psi_f = YX|1\rangle$$

$$YX = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

$$YX|1\rangle = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ i \end{bmatrix} = i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = i|1\rangle$$

$$\psi_f = YX|1\rangle \Rightarrow \psi_f = i|1\rangle$$



4.. Find the final qubit state for the circuit —[Y]—[Z]— Using inputs $|0\rangle$ and $|1\rangle$

Apply Y followed by Z on $|0\rangle$

$|0\rangle \xrightarrow{Y} \xrightarrow{Z}$

$\psi_i = |0\rangle$
 $\psi_f = ZY|0\rangle$

$ZY|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} |0\rangle$

$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} |0\rangle$

$\psi_f = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\psi_f = \begin{bmatrix} 0 \\ -i \end{bmatrix} \Rightarrow -i \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -i|1\rangle$

$\psi_i = |0\rangle \Rightarrow \psi_f = -i|1\rangle$

Apply Y followed by Z on $|1\rangle$

$|1\rangle \xrightarrow{Y} \xrightarrow{Z}$

$\psi_i = |1\rangle \Rightarrow \psi_f = ZY|1\rangle$

$ZY|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} |1\rangle$

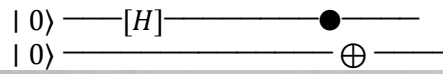
$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} |1\rangle$

$\psi_f = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -i \\ 0 \end{bmatrix} = -i \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

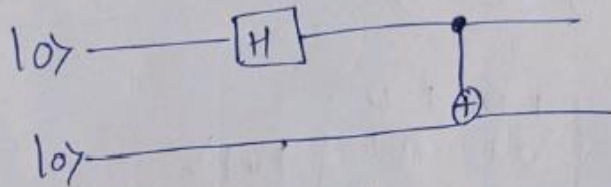
$\psi_f = -i|0\rangle$



5. Find the final state of the following circuit:



Applying H gate and CNOT gate



$$\psi_i = |00\rangle$$

A Hadamard gate is applied to the first qubit, followed by a CNOT gate.

$$\psi_f = \text{CNOT} (H \otimes I) |00\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\Psi_f = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} |00\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} |00\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

multiple with $|00\rangle$. This operation
pull out the first column of the matrix

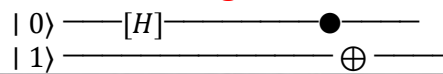
$$\Psi_f = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Psi_f = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right]$$

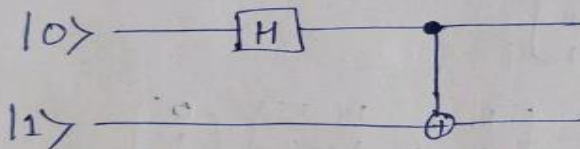
$$\Psi_f = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

This is the Entangled State or Bell state.

6. Find the final state of the following circuit:



Entanglement using Quantum Circuits



The system starts with two separate qubits

→ A Hadamard gate (H) is applied to the first qubit $|0\rangle$.

→ The second qubit $|1\rangle$ is unchanged.



$$\psi_i = |01\rangle$$

$$\psi_f = \text{CNOT}(H \otimes I) |01\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\psi_f = \text{CNOT} \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) |01\rangle$$

$$= \text{CNOT} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$\psi_f = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

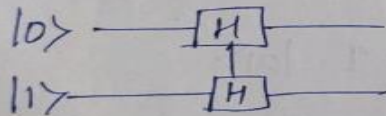
This is a Entangled State known as a Bell State.



7. Find the final state of the following circuit:

$$\begin{array}{l} |0\rangle \text{ --- } [H] \text{ ---} \\ |1\rangle \text{ --- } [H] \text{ ---} \end{array}$$

Applying a Hadamard (H) gate to Each qubit in a 2-qubit System



A Hadamard gate is applied to the first qubit, and another H-gate is applied to the second qubit. This parallel operation is represented by the tensor product $(H \otimes H)$

$$\psi_i = |01\rangle$$

$$\psi_f = (H \otimes H) |01\rangle$$

Using matrix mechanics

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H \otimes H = \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \otimes \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right)$$

$$= \frac{1}{2} \begin{bmatrix} 1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} & 1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ 1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} & -1 \times \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

$$(H \otimes H) |01\rangle = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

The vector $|01\rangle$ only has a '1' in the second position, this operation simply pulls out the second column of the matrix.

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$|11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



$$\psi_f = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$
$$\psi_f = \frac{1}{2} \left[\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right]$$

Converting each vector back to its ket representation

$$\psi_f = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

So both qubits together are in superposition of four states. $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

8 . Find the final state of the following circuit:

$$\begin{array}{c} |0\rangle \text{ --- [H] --- } \\ |1\rangle \text{ --- } \end{array}$$

Superposition using Quantum Circuit :-

Superposition is a fundamental property of quantum mechanics where a qubit can exist in a combination of basis states $|0\rangle$ and $|1\rangle$ simultaneously.

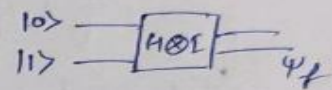
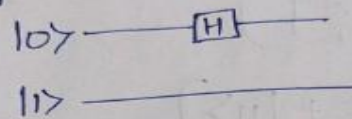


A Quantum Circuit can be used to create Superposition using the Hadamard gate (H gate).

Consider a 2 qubit system initially in the state $|\psi_i\rangle = |01\rangle$

A Hadamard gate is applied only on the first qubit and second qubit is left unchanged.

Circuit diagram.



$$\psi_f = (H \otimes I) (|0\rangle \otimes |1\rangle)$$

$$\psi_f = (H \otimes I) |01\rangle$$

$$\text{Matrices } H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$H \otimes I |01\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes I |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \times I & 1 \times I \\ 1 \times I & -1 \times I \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} |01\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$\psi_f = \frac{1}{\sqrt{2}} (|01\rangle + |11\rangle)$$

So both qubits together are in a superposition of two states $|01\rangle$ and $|11\rangle$



9. Explain the difference between classical circuits and quantum circuits?

<u>Classical Circuits</u>	<u>Quantum Circuits</u>
① Basic unit of Classical is Bit (0 or 1)	① Basic unit of Quantum is Qubit (Superposition of 0 and 1)
② State representation of Classical single definite value (0 or 1)	② State representation of Quantum is vector in Hilbert space E : Eg: $\alpha 0\rangle + \beta 1\rangle$
③ Classical operate based on Boolean algebra and deterministic logic gates AND, OR etc.	③ Quantum Linear algebra matrices, tensors, vectors, Unitary operators.
④ Classical most gates are irreversible	④ Quantum All gates must be reversible.
⑤ Classical single input and multiple outputs	⑤ Quantum single input and single output.
⑥ Classical No entanglement	⑥ Quantum can be entangled, creating correlations not possible classically.

UNIT V:

Short Question & Answers:

1. Write two applications of quantum computing in cybersecurity

1. Quantum Key Distribution (QKD)

- QKD uses photons to send secret keys securely.
- Any attempt to hack the communication immediately changes the photons, alerting users.

2. Fraud Detection

- Quantum machine learning can detect unusual patterns in financial transactions.



- This helps banks identify fraud faster and more accurately.

2. Mention any two applications of quantum deep learning.

1. Improved Optimization

- Deep learning needs to find the best parameters for accuracy.
- Quantum algorithms help reach these best values faster than classical gradient descent.

2. Big Data Handling

- Quantum parallelism allows quick processing of huge amounts of data.
- This makes training large deep learning models more efficient.

3. Write two ethical applications/concerns in quantum computing.

1. Breaking Current Encryption

- Quantum computers can break existing internet security methods.
- This may make current online data and communications unsafe.

2. Privacy Risks

- Sensitive personal and financial data could be exposed if quantum technology is misused.
- This increases the risk of identity theft and fraud.

4. Mention two applications of quantum artificial intelligence

1. Faster ML Training

- Quantum computers reduce the training time for AI and machine learning models.
- This helps develop AI applications more quickly.

2. Better Search Algorithms

- Grover's algorithm allows faster search through large databases.
- This improves efficiency in AI-based data retrieval.

5. Write two applications of quantum computing in molecular modelling

1. Drug Discovery

- Quantum models study how drug molecules interact with proteins.
- This reduces long lab experiments and speeds up new medicine development.

2. Simulating Chemical Reactions

- Quantum computers can model complex reactions faster than classical computers.
- This helps chemists understand reaction pathways clearly.

6. Describe any two applications of Quantum Neural Networks (QNNs) in real-world problem-solving

1. Medical Diagnosis

- QNNs can analyze medical scans and patient data faster.
- This helps doctors identify diseases more quickly and accurately.

2. Fraud Detection

- QNNs detect unusual or suspicious transactions in real time.
- This helps banks prevent financial fraud effectively.

7. What does “post-quantum cryptography” mean?

Post-quantum cryptography

Post-quantum cryptography refers to cryptographic techniques designed to remain secure even when quantum computers are able to break current encryption methods.



8.Name one quantum algorithm used for Quantum Speed-up for Data Search

Grover's Algorithm is a quantum algorithm that provides faster searching of unsorted databases compared to classical algorithms.

9.What is the role of quantum computing in secure authentication systems?

Quantum computing plays a role in secure authentication by enabling stronger security methods based on quantum principles that are difficult to break.

10.How does quantum computing improve secure authentication systems

Quantum computing improves secure authentication by detecting eavesdropping and preventing duplication using quantum properties like no-cloning.

Long Question & Answers:

1.Explain the applications of quantum computing in artificial intelligence

1. Faster ML Training

- Quantum computers reduce the training time for AI and machine learning models.
- This helps develop AI applications more quickly.

2. Better Search Algorithms

- Grover's algorithm allows faster search through large databases.
- This improves efficiency in AI-based data retrieval.

3. Big Data Analytics

- Quantum AI can process and analyze huge datasets easily.
- This helps in identifying trends and patterns faster.

4. Quantum Robots

- Robots with quantum processors can make smarter and faster decisions.
- This improves their performance in complex environments.

5. Predictive Healthcare

- Quantum AI can detect diseases at an early stage.
- This supports better treatment and preventive care.

6. Optimization (Logistics & Traffic)

- Quantum AI helps in planning routes to reduce travel time and fuel usage.
- It also helps manage traffic congestion efficiently.

7. Climate Modelling

- Quantum AI predicts weather and climate patterns with high accuracy.
- This helps in disaster preparedness and environmental planning.

8. Real-Time Language Translation

- Quantum AI enables faster and more accurate language translation.
- This improves communication in global applications.

9. Autonomous Drones

- Quantum AI helps drones navigate safely and avoid obstacles.
- This improves efficiency in delivery and surveillance tasks.



10. Smart City Systems

- Quantum AI helps manage electricity, traffic, and surveillance in cities.
- This leads to better control and more efficient urban living.

2.Explain the major applications of quantum computing in molecular modelling and new material design.

Molecular Modelling Applications:

1. Predicting Molecular Structures

- Quantum computers simulate atoms and electrons with high accuracy.
- This helps scientists understand the exact shape and behavior of molecules.

2. Drug Discovery

- Quantum models study how drug molecules interact with proteins.
- This reduces long lab experiments and speeds up new medicine development.

3. Simulating Chemical Reactions

- Quantum computers can model complex reactions faster than classical computers.
- This helps chemists understand reaction pathways clearly.

4. Protein Folding Studies

- Quantum systems can predict protein shapes accurately.
- This is useful for understanding diseases caused by misfolded proteins.

New Material Design Applications:

5. Designing Better Batteries

- Quantum computing helps find materials that store more energy.
- It supports the development of long-lasting and fast-charging batteries.

6. Superconductor Design

- Quantum computers predict materials that conduct electricity with zero resistance.
- This helps in creating efficient power and electronic systems.

7. New Polymer Materials

- Quantum models help design strong, lightweight polymer materials.
- These materials are useful in aerospace and engineering industries.

8. Catalyst Optimization

- Quantum simulations help industries find catalysts that speed up reactions.
- This reduces energy use and improves production efficiency.

9. Predicting Material Properties

- Quantum computers can predict properties like magnetism and conductivity.
- This helps scientists design materials for specific uses.

10. Carbon Capture Materials

- Quantum models help find materials that absorb CO₂ more effectively.
- This supports climate control and pollution-reduction efforts.

3.Explain the role of quantum computing in improving cybersecurity applications

1. Post-Quantum Cryptography

- New encryption methods are being developed to resist quantum computer attacks.
- These methods keep data safe even when powerful quantum systems appear.

2. Breaking Classical Encryption

- Quantum algorithms like Shor's can break RSA and ECC very quickly.
- This shows that current classical encryption is no longer fully safe.

3. Quantum Key Distribution (QKD)

- QKD uses photons to send secret keys securely.
- Any attempt to hack the communication immediately changes the photons, alerting users.



4. Fraud Detection

- Quantum machine learning can detect unusual patterns in financial transactions.
- This helps banks identify fraud faster and more accurately.

5. Protecting Critical Systems

- Quantum security protects government, military, and power grid networks.
- This keeps important national systems safe from cyber-attacks.

6. Quantum-Secure Cloud

- Data stored in the cloud becomes extremely hard to hack with quantum encryption.
- This ensures high-level protection for sensitive information.

7. Blockchain Security

- Quantum hashing strengthens blockchain networks.
- This makes digital currencies and records more secure.

8. Intrusion Detection

- Quantum systems can analyze network traffic in real time.
- This allows quick detection of cyber-attacks and unusual activities.

9. Secure Transactions

- Quantum security ensures safety during bank transfers and online payments.
- This reduces the chances of hacking and financial theft.

10. Authentication Systems

- Quantum randomness helps create passwords that are impossible to guess.
- This improves login security for devices and online accounts.

4. Discuss the applications of quantum computing in deep learning

1. Faster Training of Deep Models

- Quantum computers can process many possibilities at the same time.
- This helps deep learning models train much faster than on normal computers.

2. Improved Optimization

- Deep learning needs to find the best parameters for accuracy.
- Quantum algorithms help reach these best values faster than classical gradient descent.

3. Big Data Handling

- Quantum parallelism allows quick processing of huge amounts of data.
- This makes training large deep learning models more efficient.

4. Better Image Classification

- Quantum algorithms recognize patterns more clearly.
- This improves the accuracy of identifying images and objects.

5. Enhanced NLP Models

- Natural Language Processing becomes faster and more accurate with quantum computing.
- It helps in better translation, summarization, and text understanding.

6. Quantum Reinforcement Learning

- Quantum systems help agents make faster decisions.
- This is useful for robots and AI systems that learn from their environment.

7. Recommendation Systems

- Quantum models quickly analyze user preferences.
- This improves suggestions on platforms like YouTube, Netflix, or Amazon.

8. Medical Imaging

- Deep quantum models can detect tumors and diseases faster.
- This helps doctors diagnose illnesses more accurately.

9. Financial Forecasting

- Quantum computing finds complex patterns in market data.
- This helps make better predictions for stocks and investments.

10. Robotics



- Quantum deep learning helps robots understand their surroundings in real time.
- This improves their ability to learn tasks and adapt quickly.

5. Describe the applications of Quantum Neural Networks (QNNs) in real-world problem-solving.

1. Pattern Recognition

- QNNs can detect hidden patterns in data quickly.
- This is useful for analyzing images and identifying trends.

2. Image & Speech Recognition

- Quantum states allow QNNs to process large data efficiently.
- This improves the speed and accuracy of recognizing images and speech.

3. Medical Diagnosis

- QNNs can analyze medical scans and patient data faster.
- This helps doctors identify diseases more quickly and accurately.

4. Fraud Detection

- QNNs detect unusual or suspicious transactions in real time.
- This helps banks prevent financial fraud effectively.

5. Weather Prediction

- QNNs can handle large amounts of climate data.
- This helps predict weather patterns and extreme events more accurately.

6. Autonomous Driving

- QNNs help self-driving cars make decisions quickly.
- This improves safety and efficiency on the roads.

7. Supply Chain Optimization

- QNNs predict delivery routes and plan logistics efficiently.
- This reduces delays and improves overall supply chain management.

8. Language Understanding

- QNNs improve translation, chatbots, and virtual assistants.
- This helps machines understand and respond to human language better.

9. Robotics Control

- QNNs enhance robot learning and movement control.
- This allows robots to perform tasks more accurately and safely.

10. High-Speed Classification

- QNNs can sort and categorize large data sets extremely quickly.
- This is useful for data analysis in research, business, and AI applications.

6. Write ethical applications/concerns in quantum computing.

1. Breaking Current Encryption

- Quantum computers can break existing internet security methods.
- This may make current online data and communications unsafe.

2. Privacy Risks

- Sensitive personal and financial data could be exposed if quantum technology is misused.
- This increases the risk of identity theft and fraud.

3. Power Imbalance

- Only wealthy countries or companies may control quantum computing.
- This could create inequality in technology access and global power.

4. Job Loss

- Automation using quantum computing may reduce classical computing jobs.
- Some traditional IT and data-processing roles may become obsolete.

5. Unregulated AI Behavior



- Quantum AI systems could make biased or harmful decisions.
- Without proper rules, this can affect society negatively.

6. Military Misuse

- Quantum computing could be used for advanced warfare or spying.
- This raises serious security and ethical concerns globally.

7. Data Misuse

- Sensitive information may be abused without proper laws.
- This includes personal, medical, or financial data exploitation.

8. Bias in Quantum Machine Learning

- Quantum ML algorithms can inherit biases from their training data.
- This may lead to unfair decisions in applications like hiring or loans.

9. Health and Genetic Research Risks

- Quantum simulations could be misused for unethical medical or genetic experiments.
- This raises moral and legal concerns about human rights.

10. Environmental Impact

- Quantum computers require cooling systems that consume a lot of energy.
- This can negatively affect the environment if not managed sustainably.